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Combating counterfeits with the Blockchain-technology-supported platform under government enforcement

Chao He [a], Chunqiao Tan [b,*], W.H. Ip [c,d], C.H. Wu [e]

[a] School of Business, Central South University, Changsha, 410083, China
[b] School of Business, Nanjing Audit University, Nanjing, 211815, China
[c] Department of Mechanical Engineering, University of Saskatchewan, Saskatoon, Canada
[d] Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, Hong Kong, China
[e] Department of Supply Chain and Information Management, The Hang Seng University of Hong Kong, Hong Kong, China

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ABSTRACT

The intrinsic value of brand names has led to the widespread presence of counterfeits in multi-industries. Although government enforcement plays a role in combating counterfeits, it is difficult to fundamentally curb counterfeiting. This paper examines how the brand-name firm adopts the blockchain-technology-supported (BTS) platform in dealing with counterfeits under government enforcement. We consider a market with a manufacturer (i.e., a brand-name firm) and a counterfeiter. First, we develop a signaling game model to analyze the negative impact of deceptive sales. The results show that, under weak government enforcement and a high rampancy level of counterfeits, deceptive sales will decrease the expected demand for authentic products, the manufacturer's expected profit, the expected consumer surplus, and the expected social welfare. Although government enforcement forces the counterfeiter to implement non-deceptive sales, it will decrease the expected consumer surplus under symmetric quality information. Then, we develop a Stackelberg game model to analyze the manufacturer's adoption strategy for the BTS platform. We find that weaker government enforcement will incentivize the manufacturer to adopt the BTS platform. However, the manufacturer does not always adopt the BTS platform even if the counterfeiter implements deceptive sales. Finally, we analyze the effect of adopting the BTS platform under government enforcement. We reveal that the role of the BTS platform in improving consumer rights is stronger than that in combating counterfeits, and the effect of adopting the BTS platform will strengthen as government enforcement weakens.

1. Introduction

The brand name brings consumers a kind of symbolic meaning of buying high quality and perfect function, which allows brand-name firms to gain huge market shares and economic benefits (Green and Smith, 2002). Therefore, the brand name is the most significant intangible asset of many firms. It is estimated that Nike's brand name is worth up to $33.2 billion in 2022 (Brand Finance, 2022). Unfortunately, the intrinsic value of the brand name also provides an incentive for illegal groups to imitate the authentic product, that is, counterfeiting. Since counterfeiters need not bear the research and development costs of products and can take advantage of brand name influence to grab high profits, counterfeiting takes place in almost all industries. For example, some

* Corresponding author.
E-mail address: chunqiaot@sina.com (C. Tan).

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unauthorized retailers produce and sell counterfeit Apple accessories (Lepido, 2021); some luxury brands, such as Gucci and Louis Vuitton, have been threatened by a large number of counterfeits (Cullinane, 2021); counterfeit versions of Pfizer's COVID-19 vaccine show up in many countries, e.g., Mexico and Poland (Morris, 2021). The Organization for Economic Co-operation and Development (OECD) estimates that counterfeits trade value amounted to as much as 3.3% of world trade value in 2016 (OECD, 2019).

Governments have paid close attention to the counterfeiting problem, and have taken some enforcement actions to combat counterfeits. The UK Border Agency has stepped up efforts to combat counterfeits through cross-border cooperation with the police, customs and intellectual property protection institutions, and implemented strict monitoring at major airports, ports, and mail centers to prevent counterfeits from entering the UK. They seized more than GBP 11.6 billion worth of counterfeits in 2016 (Intellectual Property Office, 2020); the U.S. Immigration and Customs Enforcement announced that more than 327,000 counterfeits were seized by U.S. and Mexican authorities in joint anti-counterfeiting action in 2019. However, on the whole, counterfeiting has not been fundamentally curbed. The IPO (Intellectual Property Office) estimated the total value of counterfeits seized worldwide in 2013 was $461 billion, but this figure rose to $509 billion in 2016 (Intellectual Property Office, 2020). One of the reasons for this is that, due to the gradual expansion of the market scale, it is difficult for a government to fully supervise counterfeiting (Shepard, 2017). Another important reason is that government enforcement does not fundamentally solve the information asymmetry problem of product quality. Specifically, since consumers cannot completely distinguish all counterfeits from authentic ones, they may buy counterfeits unknowingly. This motivates illegal groups to invade the market.

To prevent counterfeiters from deceiving consumers and maintain their brand image, brand-name firms have taken some anti-counterfeiting measures. For example, they attempt to redesign the product in ways that are difficult to be easily counterfeited (Green and Smith, 2002). However, due to the development of counterfeiting technology, such anti-counterfeiting measure is not enough to effectively solve the counterfeiting problem (e.g., some brand-name products, such as Adidas and Uniqlo, are produced by original equipment manufacturers who may privately produce counterfeits). In addition, some information technologies are also applied in dealing with counterfeits, such as radio frequency identification (RFID) technology. However, since the information stored in the RFID tags may be altered or copied in the process of product circulation, consumers may be concerned about the product information processed by RFID technology (Pun et al., 2021; Shen et al., 2021a).

The emergence of blockchain technology provides an effective solution for eliminating asymmetric information. Blockchain is essentially a distributed data chain in which each block records the transaction information. Due to its nature of decentralization, the data recorded in the blockchain cannot be tampered with by individuals, and anyone can trace back to the source to query and verify the authenticity of these data (Choi and Luo, 2019; Dutta et al., 2020). Blockchain technology has been successfully implemented in many fields including financial transactions (Aghamohammadzadeh and Valilai, 2020; Yoon et al., 2020), the Internet of Things (IoT) (Banerjee et al., 2018, Liu et al., 2021), the manufacturing industry (Choi, 2019), the energy sector (Dong et al., 2022), the food safety (Casino et al., 2020; Wu et al., 2021) and the healthcare (Khezr et al., 2019). In addition, since counterfeiters cannot provide traceable product information, blockchain technology also can be used to combat counterfeits.

To integrate product circulation information from production to sales and lower the investment in blockchain infrastructure, many blockchain-technology-supported (BTS) platforms have been built (Shen et al., 2021a). The BTS platform connects all authorized firms to a transparent shared database. By cooperating with authorized retailers, brand-name firms can provide consumers with traceable product information from production to sales. At present, some brand-name firms have cooperated with BTS retailers to adopt the BTS platform for dealing with counterfeiting. For example, the Richline Group (a famous jewelry manufacturer) and Helzberg (a US jewelry retailer) have jointly adopted the BTS platform TrustChain to improve the traceability of diamonds, gemstones, and jewelry (Hill, 2018); some famous food manufacturers (e.g., Dole, Unilever and Tyson Foods) have teamed up with Walmart to ensure food safety by adopting the BTS platform Food Trust (Babich and Hilary, 2020). When shopping, consumers can easily verify all the details of the authentic product by scanning the QR codes on the package. Therefore, adopting the BTS platform can realize the quality disclosure effect. In practice, however, only a subset of brand-name firms adopts the BTS platform in their operation. For example, Prada, LVMH, and Cartier jointly establish the BTS platform AURA to trace the source of luxury goods in 2021. Up to now, only a few brands (e.g., Italian luxury group OTB) have chosen to adopt AURA (Ledger Insights, 2021). One important reason is that the lack of coordinated pricing with BTS retailers may damage the profits of brand-name firms (Shen et al., 2021a). That is, the indirect sales through BTS retailers lead to the double marginalization effect, which is not beneficial to brand-name firms. In addition, the adopting strategy for the BTS platform is also greatly affected by government enforcement. This is because the government combating level will affect the profits of brand-name firms, but the impact may change when the BTS platform is adopted. However, the impact of government enforcement on BTS platform adoption has not been investigated.

Motivated by this research gap, our paper discusses the application of the BTS platform in combating counterfeits under government enforcement. We seek to answer the following research questions from three perspectives:

(1) The importance of adopting the BTS platform: Without the BTS platform, what is the negative impact of deceptive sales implemented by counterfeiters?
(2) The strategy of adopting the BTS platform: What is the manufacturer's adoption strategy for the BTS platform? And how does government enforcement affect the strategy above?
(3) The value of adopting the BTS platform: What is the effect of adopting the BTS platform? And how does government enforcement affect the result above?

We answer the research questions with game-theoretic analytical models. Considering a market with a manufacturer (i.e., brand-name firm) and a counterfeiter. Under government enforcement, the counterfeiter may be expelled from the market, depending on the

government combating level. If the manufacturer does not adopt the BTS platform, the counterfeiter may deceive consumers by setting the same price as the authentic product (i.e., implementing deceptive sales); if the manufacturer adopts the BTS platform, any consumer can identify counterfeits. The research framework of our paper is as follows. First, we develop a novel signaling game model under the scenario without the BTS platform. Based on the separating equilibrium outcome and the pooling equilibrium outcome, we analyze the negative impact of deceptive sales on the market and consumers. Then, we develop a Stackelberg game model under the scenario with the BTS platform. By comparing the manufacturer's expected profits under the two scenarios, we derive the manufacturer's adopting strategy for the BTS platform under government enforcement. Finally, based on the manufacturer's adopting strategy for the BTS platform, we examine the effect of adopting the BTS platform under government enforcement.

Our major contributions are summarized as follows. First, we highlight the limitations of government enforcement in addressing counterfeiting. Specifically, a higher combating level will decrease the expected consumer surplus under symmetric quality information. Second, we reveal the impact of government enforcement on the manufacturer's adoption strategy for the BTS platform. Specifically, weaker government enforcement will incentivize the manufacturer to adopt the BTS platform. Third, we systematically analyze the effect of adopting the BTS platform under government enforcement. Specifically, we point out that the role of the BTS platform in improving consumer rights is stronger than that in combating counterfeits, and the effect of adopting the BTS platform will strengthen as government enforcement weakens. Our findings can help brand-name firms to better develop the adoption strategy for the BTS platform.

The rest of this paper is organized as follows. In Section 2, we review related literature. In Section 3, we give the problem description. In Section 4, we develop a signaling game model and a Stackelberg game model under the scenarios with and without the BTS platform, respectively. Based on the equilibrium outcomes under different scenarios, we analyze the negative impact of deceptive sales, and derive the manufacturer's adopting strategy for the BTS platform. In Section 5, we examine the effect of adopting the BTS platform under government enforcement. In section 6, we check the robustness of our results. We conclude this paper in section 7. All the proofs are provided in the Appendix A.

2. Literature review

Summarizing the existing literature, most studies on counterfeits are empirical studies that mainly focus on analyzing consumers' intentions to purchase counterfeits. Kim et al. (2016) present some important factors affecting consumers' purchase intentions, such as level of legal knowledge, purchase experience of authentic products and counterfeits. Chen et al. (2018) apply the moral disengagement theory to analyze the intentions of consumers to purchase counterfeit luxuries. The empirical results demonstrate that consumers' desire to purchase counterfeit luxuries lies in moral rationalization and moral decoupling. Gultekin (2018) examines the impact of the love of money on consumers' intentions to purchase counterfeit apparel. The questionnaire results show that there is a significant positive correlation between them. Davidson et al. (2019) analyze the mediating factors for the correlation between materialism and consumers' purchase intentions for counterfeits, and summarize these factors as the risk of embarrassment, counterfeit detectability and product conspicuousness. Different from these empirical papers, we examine the counterfeiting problem using an analytical method. Below, we sort out some analytical studies on counterfeits and classified them according to the types of counterfeits[1].

One stream of analytic studies focuses on non-deceptive counterfeits. Zhang et al. (2012) analyze the impact of non-deceptive counterfeits on the price and demand of authentic products, and how brand-name firms can combat counterfeiting. They indicate that counterfeits lower the authentic price but increase demand for authentic products. In addition, improving product quality is an effective anti-counterfeiting solution. Lahiri and Dey (2013) discuss how piracy enforcement actions affect manufacturers' investment decisions in product quality. The results show that lower piracy enforcement increases the manufacturer's incentive to invest in quality. Pun and Deyong (2017) develop a two-period game theoretical model to examine the impact of non-counterfeits on brand-name firms. Gao et al. (2017a) investigate the entry of counterfeits and show that counterfeits with a high physical resemblance but a low quality are more likely to enter the market by defying the deterrence of the brand-name firm. Gao et al. (2017b) reveal that the entry of non-deceptive counterfeits does not always reduce consumer surplus and social welfare. Yi et al. (2020) examine who among the supply chain members is in the best position to perform anti-counterfeiting activities. They reveal that the manufacturer should lead a retailer to combat counterfeits. Different from these papers, our paper focuses on deceptive counterfeits.

Another stream of analytic studies focuses on deceptive counterfeits. Qian (2014) demonstrates that deceptive counterfeits increase manufacturers' incentive to improve product quality. Qian et al. (2015) developed a novel model in which product quality has two attributes, such as searchable attribute and experiential attribute, and discuss the brand-name firm's quality improvement strategy to deal with deceptive counterfeits. They reveal that brand-name firms prefer to invest more in improving searchable quality. Cho et al. (2015) analyze the impact of counterfeit types on the effectiveness of anti-counterfeiting strategies and highlight that the anti-counterfeiting strategies which are effective for dealing with non-deceptive counterfeits may not work well for dealing with deceptive counterfeits. Zhang and Zhang (2015) examine the impact of deceptive counterfeits on the brand name company's optimal channel decision. They indicate that adopting a dual-channel structure may incentivize the retailer to sell more counterfeits, so it is not always the best choice. Sun et al. (2020) analyze the online platform's strategy to combat deceptive counterfeits and show that the

[1] Counterfeits are broadly classified into deceptive counterfeits and non-deceptive counterfeits (Grossman and Shapiro, 1988a, 1988b). Non-deceptive counterfeits refer to those that consumers can distinguish from authentic products before purchase, while those counterfeits which consumers cannot distinguish from authentic products are called deceptive counterfeits.

platform's combating intensity depends on the production cost of the authentic product.

A few scholars have also studied how to adopt blockchain technology for dealing with deceptive counterfeits. Shen et al. (2021a) examine the brand-name firm's adoption strategy for the BTS platform in a setting where there are some expert consumers. The results show that adopting the BTS platform is an effective anti-counterfeiting strategy only if the number of expert consumers is low enough. Shen et al. (2021b) develop Stackelberg game models to examine how quality inspection and blockchain adoption can battle counterfeit masks during the COVID-19 outbreak. They show that when the spread level of COVID-19 is relatively low, authentic mask sellers will adopt blockchain technology rather than relying on the government for quality inspection. Pun et al. (2021) develop a signaling game model to examine the brand-name firm's adoption strategy under the incorporation of consumers' privacy concerns. Unlike Shen et al. (2021a) and Shen et al. (2021b), we incorporate the price signaling effect (Spence, 1973) into the model. Specifically, if the manufacturer does not adopt the BTS platform, the counterfeiter can choose to implement non-deceptive sales or deceptive sales. Correspondingly, we develop a signaling game model rather than a Stackelberg game model under the scenario with the BTS platform. Although Pun et al. (2021) also apply signaling game theory to develop a model under the scenario without blockchain technology, they aim to examine the brand-name firm's strategy of self-implementing blockchain technology rather than adopting the BTS platform.

Many scholars have conducted research on how government enforcement affects the decision-making of firms. Hafezi and Zolfagharinia (2018) point out that strict government enforcement may cause firms to be reluctant to innovate and only produce a single standard product. Zhang and Yousaf (2020) illustrate that the firms' green improvement degree is influenced by government intervention, and stronger government intervention may not always lead to higher green improvement. Zheng et al. (2022) reveal that only a high regulatory punishment will drive the upstream and downstream firms of the green supply chain to reduce carbon emissions faster. Wang et al. (2022) show that strict supervision and severe punishment measures can drive collectors and recyclers to participate in the collection and recycling of waste electrical and electronic equipment. However, in the aforementioned literature on adopting blockchain technology to combat counterfeits, the impact of government enforcement on firms' adoption decisions has not been systematically discussed.

The positioning of our paper in the prior literature on counterfeits is shown in Table 1. In summary, few scholars use signaling game theory to discuss how to adopt the BTS platform to combat deceptive counterfeits and add government enforcement to the model. Our paper expands the relevant literature stream. Specifically, we consider the role of government enforcement and develop a novel signaling game model under the scenario without the BTS platform. By comparing the equilibrium outcomes under the scenarios with and without the BTS platform, we analyze the manufacturer's adopting strategy and the effect of the BTS platform. In addition, we also discuss the negative impact of deceptive sales and the impact of government enforcement on the effect of the BTS platform adoption, which fills theoretical gaps on counterfeiting.

3. Problem description

Consider a manufacturer (firm M) selling authentic products to consumers. At the same time, there is a counterfeiter (firm C) that sells counterfeits with low quality but a high physical resemblance to authentic products. Without loss of generality, the true quality of the authentic product is normalized to 1 and the true quality of the counterfeit is $q_C = q$, where $0 < q < 1$. For analytical tractability, we assume that the unit production cost of the authentic product is zero. This assumption will be relaxed in the robustness check. We adopt the vertical differentiation model to describe consumers' utilities for different products as $U_i(\theta) = \theta\phi_i - p_i$, where θ represents consumers' taste in quality; p_i is the price of the product i; ϕ_i is consumers' perceived quality for the product i. We capture customers' heterogeneity and assume that θ is uniformly distributed on [0, 1]. Each consumer's purchasing strategy depends on the utilities for different products and purchases at most one unit of a product.

If the manufacturer does not adopt the BTS platform, the counterfeiter can choose to implement deceptive or non-deceptive sales (i.e., it sets the same or different price of the counterfeit as the authentic price). When the counterfeiter implements deceptive sales (i.e., $p_C = p_M$), consumers cannot identify counterfeits by product prices. In this case, consumers are deceived and do not aware of the existence of counterfeits, thus their perceived quality for both products is equal to the true quality of the authentic product (i.e., $\phi_M = \phi_C = 1$)[2]. Let $\lambda \in (0,1)$denote the rampancy level of counterfeits in the market, representing the proportion of counterfeits among all products (Cho et al. 2015). Under deceptive sales, if a consumer chooses to purchase an authentic product, the probability that he will purchase a counterfeit is λ. When the counterfeiter implements non-deceptive sales (i.e., $p_C \neq p_M$), consumers will update their belief about the product's quality after observing all product prices (i.e., consumers can identify counterfeits by product prices). In this case, consumers' perceived quality for both products is equal to true quality (i.e., $\phi_M = 1, \phi_C = q$).

If the manufacturer adopts the BTS platform, the manufacturer sells authentic products through a BTS retailer. Let w and denote the wholesale price of the authentic product. Based on traceable product information, consumers can distinguish the counterfeit from the authentic product, so the counterfeiter only implements non-deceptive sales (i.e., $p_C \neq p_M$).

In addition, our model captures the feature that the counterfeiter may be expelled from the market under government enforcement.

[2] Different from Pun et al. (2021), we do not assume that the consumers' perceived quality for both products is the weighted average of the quality of the two types of products under deceptive sales. The reason is that it is not reasonable for consumers to know the quality of counterfeits while they are unable to identify counterfeits. Due to the development of counterfeiting technology, we believe that it is difficult for consumers to realize the existence of counterfeits under deceptive sales. Therefore, we assume that consumers' perceived quality for both products is 1 under deceptive sales. This assumption is more in line with the judgment of consumers in reality and was also adopted by Shen et al. (2021b).

Table 1
The positioning of our paper in the prior literature on counterfeits.

Paper	Non-deceptive counterfeit	Deceptive counterfeit	Signaling game	Role of government enforcement	Adoption of blockchain technology/ BTS platform
Zhang et al. (2012); Lahiri and Dey (2013); Pun and Deyong (2017); Gao et al. (2017a, 2017b); Yi et al. (2020).	√				
Qian (2014); Cho et al. (2015); Zhang and Zhang (2015); Sun et al. (2020)		√			
Qian et al. (2015)		√	√		
Shen et al. (2021a); Shen et al. (2021b)		√			√
Pun et al. (2021)		√	√		√
Our paper		√	√	√	√

Let e denote the government combating level, where $0 \le e \le 1$. That is, with probability e, the counterfeiter is caught and the manufacturer has a monopoly in the market. We first assume that the government combating level is independent of the rampancy level of the counterfeits. The reason is that it is difficult for the government to fully monitor the activity of counterfeiters in the market. Following Cho et al. (2015), we then assume that the government combating level is exogenously given. This is because the combating level as a long-term strategy cannot be changed in the short term. Table 2 shows the major notation used in our paper.

The sequence of events is as follows:

1. Under a certain government combating level, the manufacturer decides whether to adopt the BTS platform. If the BTS platform is not adopted, the manufacturer sets the authentic price p_M, otherwise the manufacturer sets the wholesale price of the authentic product w and the BTS retailer sets the authentic price p_M.
2. After observing the authentic price, the counterfeiter decides to implement deceptive sales or non-deceptive sales, and sets the price of the counterfeit p_C. If the manufacturer adopts the BTS platform, the counterfeiter can only implement non-deceptive sales.
3. Based on the price of different products, each consumer decides purchasing strategy. The market demand and firms' profits are then realized.

The market structure under different scenarios is shown in Fig. 1.

4. The model

4.1. Without the BTS platform

Under the scenario without the BTS platform, consumers judge the authenticity of products by prices. Due to the price signaling effect (Spence, 1973), there is a signaling game between the manufacturer and the counterfeiter. When the counterfeiter implements non-deceptive sales, consumers can distinguish the counterfeit from the authentic product, and there exists a separating equilibrium. The consumer's utilities for the authentic product and the counterfeit are $U_M(\theta) = \theta - p_M$ and $U_C(\theta) = q\theta - p_C$, respectively. When the counterfeiter is not expelled from the market by the government, the consumer will choose to buy an authentic product if $U_M(\theta) > U_C(\theta)$; when the counterfeiter is expelled from the market by the government, the consumer will choose to buy an authentic product if $U_M(\theta) > 0$. To sum up, with the government combating level e, the expected demands for authentic products and counterfeits under separating equilibrium can be expressed as $E(D_M^{SE}) = (1-e)(1-\frac{p_M-p_C}{1-q}) + e(1-p_M)$ and $E(D_C^{SE}) = (1-e)\left(\frac{p_M-p_C}{1-q} - \frac{p_C}{q}\right)$, respectively. Correspondingly, the expected profits of the manufacturer and the counterfeiter under separating equilibrium are as follows:

$$E(\pi_M^{SE}) = p_M E(D_M^{SE}) = p_M[(1-e)(1-\frac{p_M-p_C}{1-q}) + e(1-p_M)] \tag{1}$$

$$E(\pi_C^{SE}) = p_C E(D_C^{SE}) = p_C(1-e)(\frac{p_M-p_C}{1-q} - \frac{p_C}{q}) \tag{2}$$

When the counterfeiter implements deceptive sales, consumers cannot distinguish the counterfeit from the authentic product, and there exists a pooling equilibrium. The consumer's utility for the authentic product (counterfeit) is $U_M = \theta - p_M$. The consumer will choose to buy an authentic product if $U_M(\theta) \rangle 0$. However, if the counterfeiter is not expelled from the market by the government, he may unknowingly buy a counterfeit with probability λ. Therefore, with the government combating level e, the expected demands for authentic products and counterfeits under pooling equilibrium can be expressed as $E(D_M^{PE}) = e(1-p_M) + (1-e)(1-\lambda)(1-p_M)$ and $E(D_C^{PE}) = \lambda(1-e)(1-p_M)$, respectively. Correspondingly, the expected profits of the manufacturer and the counterfeiter under pooling equilibrium are as follows:

$$E(\pi_M^{PE}) = p_M E(D_M^{PE}) = p_M(1+e\lambda-\lambda)(1-p_M) \tag{3}$$

Table 2
Major notation used in our paper.

Notation	Definition
i	Authentic product (M), counterfeit (C)
q_i	True quality of product i, where$q_M = 1, q_C = q \in (0,1)$
p_i	Price of product i
ϕ_i	Perceived quality of product i
θ	Consumers' taste in quality, where θ $U[0,1]$
λ	Rampancy level of counterfeits, where $\lambda \in (0,1)$
w	Wholesale price of the authentic product
e	Government combating level, where $e \in [0,1]$
SE	Separating equilibrium outcome without the BTS platform
PE	pooling equilibrium outcome without the BTS platform
B	Equilibrium outcome with the BTS platform
$E(D_i^j)$	Expected demand for product i, where $j \in \{SE, PE, B\}$
$E(\pi_M^j)$	Expected profit of the manufacturer, where $j \in \{SE, PE, B\}$
$E(\pi_C^j)$	Expected profit of the counterfeiter, where $j \in \{SE, PE, B\}$
$E(\pi_R^B)$	Expected profit of the BTS retailer
$E(CS^j)$	Expected consumer surplus,$j \in \{SE, PE, B\}$
$E(SW^j)$	Expected social welfare,$j \in \{SE, PE, B\}$

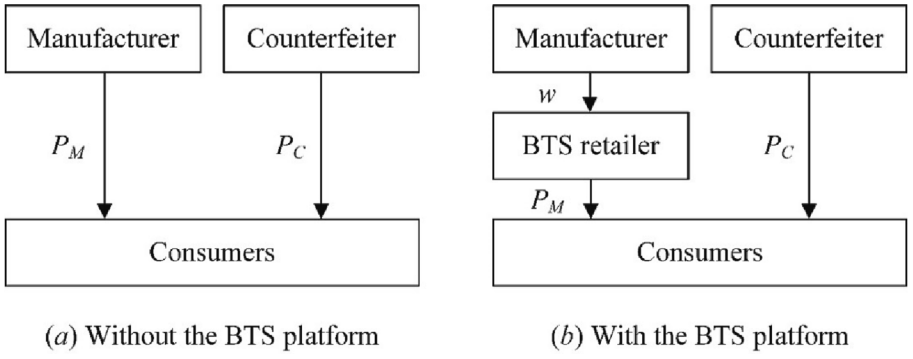


Fig. 1. The market structure under different scenarios.

$$E\left(\pi_C^{PE}\right) = p_M E(D_C^{PE}) = p_M \lambda (1-e)(1-p_M) \tag{4}$$

Which type of equilibrium outcome is obtained depends on the expected profits of the counterfeiter under different sales strategies. Specifically, under the given authentic price, there is a separating equilibrium only if the counterfeiter can obtain greater expected profit by implementing non-deceptive sales. Note that, although the manufacturer is a leader, the type of equilibrium outcome obtained is not related to the manufacturer's expected profit[3]. Therefore, there exists a separating equilibrium if and only if the following constraint is satisfied:

$$\max_{p_C} E[\pi_C^{SE}(p_C \neq p_M)] > E[\pi_C^{PE}(p_C = p_M)] \tag{5}$$

In this paper, we adopt intuitive criterion (Cho and Kreps, 1987) to refine the multiple equilibrium outcomes. Specifically, if the counterfeiter can benefit from a certain non-equilibrium path given the out-of-equilibrium beliefs, the intuitive criterion requires that this equilibrium outcome cannot survive. According to Equations (1)–(4) and constraint (5), we can solve this signaling game by backward induction. The equilibrium outcomes under the scenario without the BTS platform are presented as Lemma 1.

Lemma 1. *Under the scenario without the BTS platform, if* $\lambda < \overline{\lambda}(e,q) = \frac{q}{4(1-eq)}$, *there exists a separating equilibrium and the prices of the authentic product and the counterfeit are* $p_M^{SE} = \frac{1-q}{2-q-eq}$ *and* $p_C^{SE} = \frac{q(1-q)}{2(2-q-eq)}$, *otherwise there exists a pooling equilibrium and the authentic price is* $p_M^{PE} = \frac{4(1-q)\lambda}{q+4\lambda-4\lambda q}$.

[3] For example, when the rampancy level of counterfeits is low enough (i.e., $\lambda \to 0$), the counterfeiter always implements non-deceptive sales (i.e., separating equilibrium) even though the manufacturer's expected profit when the counterfeiter implements non-deceptive sales is lower than when the counterfeiter implements deceptive sales. Therefore, the type of equilibrium outcome obtained is not related to the manufacturer's expected profit.

Lemma 1 shows that the counterfeiter will implement non-deceptive sales only when the rampancy level of counterfeits is lower than a certain threshold. Next, we examine the impact of deceptive sales on the expected demand for authentic products, the manufacturer's expected profit, the expected consumer surplus, and the expected social welfare. The expected consumer surplus under separating equilibrium and pooling equilibrium can be expressed as $E(CS^{SE}) = e\int_{p_M}^{1}\theta - p_M d\theta + (1-e)\int_{\frac{p_M-p_C}{1-q}}^{1}\theta - p_M d\theta + (1-e)\int_{\frac{p_C}{q}}^{\frac{p_M-p_C}{1-q}} q\theta - p_C d\theta$ and $E(CS^{PE}) = \int_{p_M}^{1}\theta - p_M d\theta$. The expected social welfare under separating equilibrium and pooling equilibrium can be expressed as $E(SW^{SE}) = E(CS^{SE}) + E(\pi_M^{SE})$ and $E(SW^{PE}) = E(CS^{PE}) + E(\pi_M^{PE})$.[4] The analysis results are presented in Proposition 1.

Proposition 1. *When the counterfeiter implements deceptive sales, then:*

(i) *the expected demand for authentic products decreases (i.e., $E(D_M^{PE}) < E(D_M^{SE})$) if $e \le e_1(q) = \frac{4-3q}{2q}$ and $\lambda > \lambda_1(e,q) = \frac{q}{2(2-q-eq)}$.*

(ii) *the manufacturer's expected profit decreases (i.e., $E(\pi_M^{PE}) < E(\pi_M^{SE})$) if $e \le e_2(q) = \frac{4-2q-\sqrt{32q-16-14q^2}}{4q}$ and $\lambda > \lambda_2(e,q) = \frac{2q-2eq^2+q\sqrt{2q(1-e)(2-q-eq)}}{4(e^2q^2-2eq+q^2-2q+2)}$.*

(iii) *the expected consumer surplus always decreases (i.e., $E(CS^{PE}) < E(CS^{SE})$). Moreover, $\frac{\partial[E(CS^{SE})-E(CS^{PE})]}{\partial e} < 0$.*

(iv) *there exist thresholds $\lambda_3(e,q)$ and $e_3(q)$ such that the expected social welfare decreases if $e \le e_3(q)$ and $\lambda > \lambda_3(e,q)$.*

With the reduction of the government's combating level, the monopoly demand for authentic products and the monopoly profit of the manufacturer will decrease. Therefore, as shown in Proposition 1(*i*) and Proposition 1(*ii*), under weak government enforcement, when the rampancy level of counterfeits is higher than a certain threshold, deceptive sales will decrease the demand for authentic products and the manufacturer's expected profit. When the counterfeiter implements deceptive sales, the manufacturer will increase the authentic price to compensate for the profit loss caused by consumers unintentionally purchasing counterfeits (i.e., $p_M^{PE} > p_M^{SE}$). This implies that the price of the counterfeit will also increase (i.e., $p_C^{PE} > p_C^{SE}$). Furthermore, the price of two types of products will increase more as the government combating level decreases (i.e., $\frac{\partial(p_M^{PE}-p_M^{SE})}{\partial e} < 0$, $\frac{\partial(p_M^{PE}-p_C^{SE})}{\partial e} < 0$). Thus, as shown in Proposition 1(*iii*), deceptive sales always decrease the expected consumer surplus, and weaker government enforcement will deepen the negative impact of deceptive sales on consumers. In addition, as deceptive sales may increase the manufacturer's expected profit, it is also possible to increase expected social welfare. However, under weak government enforcement, when the rampancy level of counterfeits is higher than a certain threshold, deceptive sales still decrease the expected social welfare.

To sum up, Proposition 1 reveals that, under weak government enforcement and a higher rampancy level of counterfeits, deceptive sales will decrease the expected demand for authentic products, the manufacturer's expected profit, the expected consumer surplus, and expected social welfare. These results imply that it is quite necessary to eliminate asymmetric quality information. Next, we analyze the role of government enforcement in dealing with asymmetric quality information, as shown in Proposition 2.

Proposition 2. *Although a higher combating level forces the counterfeiter to implement non-deceptive sales, it will decrease the expected consumer surplus under symmetric quality information.*

In the case of a monopoly, the manufacturer will choose to set a high price to maximize profit. Moreover, the possibility that the manufacturer has a monopoly increases with the government combating level. Therefore, a higher combating level incentivizes the manufacturer to increase the authentic price (i.e., $\frac{\partial p_M^{SE}}{\partial e} > 0$). Faced with a higher authentic price, it is more profitable for the counterfeiter to implement the low-price strategy (i.e., non-deceptive sales). Therefore, the counterfeiter is more likely to implement non-deceptive sales under a higher government combating level. However, a higher combating level also incentivizes the counterfeiter to increase the price of the counterfeits under non-deceptive sales (i.e., $\frac{\partial p_C^{SE}}{\partial e} > 0$). Therefore, the expected consumer surplus under symmetric quality information will decrease with the government combating level. This result implies that government enforcement has a limitation to solve the problems caused by counterfeiting. Specifically, although government enforcement can help firms eliminate asymmetric quality information, it is detrimental to consumers' interests.

4.2. With the BTS platform

If the manufacturer adopts the BTS platform, consumers can distinguish the counterfeit from the authentic product and the counterfeiter only implements non-deceptive sales. The manufacturer sells authentic products through a BTS retailer, there is a Stackelberg game between the manufacturer, the BTS retailer and the counterfeiter, and the expected profits of the manufacturer, the BTS retailer and the counterfeiter are as follows:

[4] Due to the illegality of counterfeit trade, it is more meaningful to analyze the expected social welfare without the counterfeiter's expected profit. Thus, our paper does not incorporate the counterfeiter's expected profit into the expected social welfare.

$$E(\pi_M^B) = wE(D_M^B) = w\left[(1-e)(1-\frac{p_M-p_C}{1-q}) + e(1-p_M)\right] \tag{6}$$

$$E(\pi_R^B) = (p_M - w)E(D_M^B) = (p_M - w)\left[(1-e)(1-\frac{p_M-p_C}{1-q}) + e(1-p_M)\right] \tag{7}$$

$$E(\pi_C^B) = p_C E(D_C^B) = p_C(1-e)(\frac{p_M-p_C}{1-q} - \frac{p_C}{q}) \tag{8}$$

By backward induction, we can obtain the equilibrium outcome under the scenario with the BTS platform as shown in Lemma 2.

Lemma 2. *Under the scenario with the BTS platform, the prices of the authentic product and the counterfeit are* $p_M^B = \frac{3(1-q)}{2(2-q-eq)}$ *and* $p_C^B = \frac{3q(1-q)}{4(2-q-eq)}$, *the wholesale price of the authentic product is* $w = \frac{1-q}{(2-q-eq)}$.

Based on Lemma 1 and Lemma 2, we can calculate the expected demand for authentic products under separating equilibrium and the scenario with the BTS platform as $D_M^{SE} = \frac{1}{2}$ and $D_M^B = \frac{1}{4}$. These results imply that, when the counterfeiter implements non-deceptive sales, adopting the BTS platform will decrease the expected demand for authentic products. This is because, when the manufacturer adopts the BTS platform, the prices of both the authentic product and the counterfeit increase (i.e., $p_M^B > p_M^{SE}, p_C^B > p_C^{SE}$). However, since the manufacturer is a leader, the double marginalization effect has a greater impact on the authentic price (i.e., $p_M^B - p_M^{SE} > p_C^B - p_C^{SE}$), which makes more consumers purchase a counterfeit rather than an authentic product (i.e., $E(D_C^B) > E(D_C^{SE})$). In addition, it is easy to find that the wholesale price of the authentic product is equal to the authentic price under separating equilibrium (i.e., $w = p_M^{SE}$). Therefore, the manufacturer will not adopt the BTS platform when the counterfeiter implements non-deceptive sales (i.e., $\lambda < \overline{\lambda}(e,q)$). By comparing the manufacturer's expected profits under pooling equilibrium and the scenario with the BTS platform, we can determine the adoption strategy for the BTS platform, as shown in Proposition 3.

Proposition 3. *The manufacturer adopts the BTS platform if* $\lambda \geq \lambda^*(e,q) = \frac{q(3-q-2eq-\sqrt{(4-q-3eq)(2-q-eq)})}{4(1-eq)^2}$ *and* $e \leq e_4(q) = \frac{4-2q-\sqrt{q(8-5q)}}{4q}$. *Moreover,* $\lambda^*(e,q) < \overline{\lambda}(e,q)$ *and* $\frac{\partial \lambda^*(e,q)}{\partial e} > 0$.

Proposition 3 indicates that the manufacturer should adopt the BTS platform only if the rampant level of counterfeits is higher than a certain threshold and the government combating level is lower than a certain threshold. The reason is explained as follows. The manufacturer's expected profit includes monopoly profit and non-monopoly profit. If the manufacturer adopts the BTS platform, both monopoly profit and non-monopoly profit will be negatively affected by the double marginalization effect, but the monopoly profit will not be positively affected by the quality disclosure effect. When the government combating level is relatively high, the manufacturer will obtain a higher monopoly profit, and the impact of counterfeits on the manufacturer's expected profit is relatively low. In this case, if the manufacturer chooses to adopt the BTS platform, the benefits from the quality disclosure effect cannot compensate for the losses caused by the double marginalization effect. Therefore, when the government combating level is higher than a certain threshold, the manufacturer should not adopt the BTS platform. Furthermore, even under a low government combating level (i.e., $< e_2(q)$), asymmetric quality information does not always decrease the manufacturer's expected profit (i.e., $\overline{\lambda}(e,q) < \lambda < \lambda_2(e,q)$). Therefore, the manufacturer can benefit from adopting the BTS platform only if the rampant level of counterfeits is higher than a

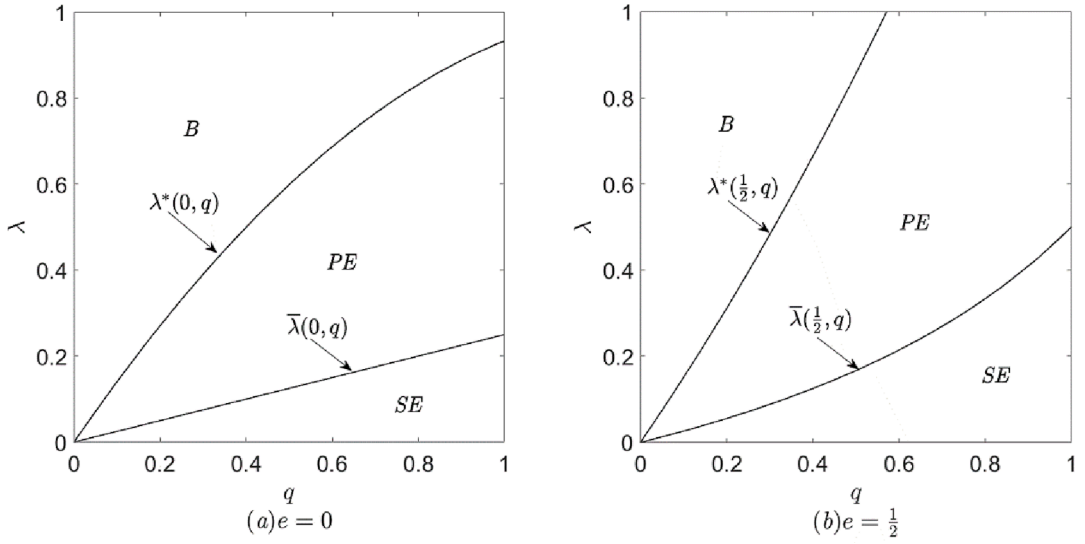


Fig. 2. The manufacturer's adoption strategies for the BTS platform.

certain threshold (i.e., $\lambda \geq \lambda^*(e, q)$). In addition, Proposition 3 also indicates that a lower combating level will incentivize the manufacturer to adopt the BTS platform. The rationale for this result lies in that a lower government combating level deepens the impact of non-monopoly profit on the manufacturer's expected profit. Accordingly, the quality disclosure effect achieved by adopting the BTS platform will be strengthened as government enforcement weakens. Proposition 3 provides a managerial implication that brand-name firms should consider the government's protection level of intellectual property rights before deciding whether to adopt the BTS platform.

Fig. 2 illustrates the manufacturer's adoption strategies for the BTS platform in cases where $e = 0$and $\frac{1}{2}$. When $e = 0$(i.e., Fig. 2(a)), the counterfeiter will implement non-deceptive sales if $\lambda < \overline{\lambda}(0, q) = \frac{q}{4}$, and the manufacturer will adopt the BTS platform if $\lambda \geq \lambda^*(0, q) = \frac{q(3-q+\sqrt{(2-q)(4-q)})}{4}$; when $e = \frac{1}{2}$(i.e., Fig. 2(b)), the counterfeiter will implement non-deceptive sales if $\lambda < \overline{\lambda}(\frac{1}{2}, q) = \frac{q}{4-2q}$, and the manufacturer will adopt the BTS platform if $\lambda \geq \lambda^*(\frac{1}{2}, q) = \frac{q(6-4q+\sqrt{(4-3q)(8-5q)})}{2(2-q)^2}$. By comparing the results above, we can find that when the government combating level increases from 0 to $\frac{1}{2}$, the counterfeiter will implement non-deceptive sales in more cases and the manufacturer will adopt the BTS platform in fewer cases. In addition, Fig. 2(b) also shows that even if the rampant level of counterfeits is high enough (i.e., $\lambda \to 1$), the manufacturer does not adopt the BTS platform when $q > \frac{4}{7}$. This is because when the quality of the counterfeit is high, adopting the BTS platform can only realize a low level of the quality disclosure effect. Therefore, if the government combating level is relatively high, the manufacturer will not choose to adopt the BTS platform.

5. Effect of adopting the BTS platform

In this section, we focus on examining the value of the BTS platform. Based on the manufacturer's adoption strategy for the BTS platform (i.e., Proposition 3), we discuss the effect of adopting the BTS platform from different perspectives and reveal some managerial insights. First, Proposition 4 summarizes the impact of adopting the BTS platform on expected demands for authentic products and counterfeits.

Proposition 4. *When the manufacturer adopts the BTS platform, then:*

(i) *the expected demand for authentic products always increases (i.e., $E(D_M^B) > E(D_M^{PE})$). Moreover, $\frac{\partial[E(D_M^B)-E(D_M^{PE})]}{\partial e} < 0$.*

(ii) *there exists a threshold $\overline{q}(e)$ such that the expected demand for counterfeits always increases (i.e., $E(D_C^B) > E(D_C^{PE})$) if $q < \overline{q}(e)$, where $\frac{\partial \overline{q}(e)}{\partial e} < 0$.*

With the BTS platform, consumers can distinguish the counterfeits from the authentic product, and the BTS retailer will set a lower authentic price (i.e., $p_M^B < p_M^{PE}$). Due to quality differences, consumers with high-quality preferences will choose to buy authentic products. Therefore, as shown in Proposition 4(*i*), adopting the BTS platform always increases the demand for authentic products. Proposition 4(*i*) also shows that, as the government combating level decreases, the improvement level in demand for authentic products from adopting the BTS platform will increase. This is because when the BTS platform is adopted, the price difference between the authentic product and the counterfeit decreases with the decrease of the government combating level (i.e., $\frac{\partial(p_M^B - p_C^B)}{\partial e} > 0$), which prompts some consumers to switch from buying counterfeits to authentic products.

Interestingly, Proposition 4(*ii*) indicates that adopting the BTS platform will also increase the expected demand for counterfeits if the quality of the counterfeit is lower than a certain threshold. The reason is explained as follows. If the quality of the counterfeit is relatively low (i.e., $q < \overline{q}(e)$), the counterfeiter will set a lower price, which attracts some consumers with low-quality preferences to buy counterfeit. Therefore, the expected demand for counterfeits will increase. Proposition 4(*ii*) also indicates that under a lower government combating level, adopting the BTS platform is more likely to decrease the demand for counterfeits. This is also because the price difference between the authentic product and the counterfeit decreases with the decrease of the government combating level.

Then, Proposition 5 summarizes the impact of adopting the BTS platform on the counterfeiter's expected profit.

Proposition 5. *When the manufacturer adopts the BTS platform, the counterfeiter's expected profit always decreases (i.e., $E(\pi_C^B) < E(\pi_C^{PE})$). Moreover, $\frac{\partial[E(\pi_C^{PE})-E(\pi_C^B)]}{\partial e} < 0$.*

Due to the quality disclosure effect, the counterfeiter has to reduce the price (i.e., $p_C^B < p_C^{PE}$). If the quality of the counterfeit is relatively high (i.e., $q \geq \overline{q}(e)$), adopting the BTS platform will decrease the expected demand for counterfeits (i.e., Proposition 4 (*i*)), so the counterfeiter's expected profit will decrease. If the quality of the counterfeit is relatively low (i.e., $q < \overline{q}(e)$), although adopting the BTS platform may increase the expected demand for counterfeits, the counterfeiter's expected profit will still decrease due to the low unit profit of counterfeits. In addition, Proposition 5(*i*) also indicates that under a lower government combating level, adopting the BTS platform can decrease the counterfeiter's expected profit to a greater extent. This is because, under weaker government enforcement, the counterfeiter will steal more non-monopoly profit from the manufacturer. This result also implies that the quality disclosure effect will strengthen as government enforcement weakens. Proposition 4 and Proposition 5 provide some important managerial insights. Specifically, adopting the BTS platform is more effective in combating high-quality counterfeits than low-quality counterfeits, and the combating effect will strengthen as government enforcement weakens.

Finally, Proposition 6 summarizes the impact of adopting the BTS platform on the expected consumer surplus and expected social

welfare.

Proposition 6. *When the manufacturer adopts the BTS platform, the expected consumer surplus and expected social welfare always increases (i.e.,* $E(CS^B) > E(CS^{PE})$, $E(SW^B) > E(SW^{PE})$*). Moreover,* $\frac{\partial[E(CS^B)-E(CS^{PE})]}{\partial e} < 0$, $\frac{\partial[E(SW^B)-E(SW^{PE})]}{\partial e} < 0$.

The expected consumer surplus depends on the difference between the true quality of the product purchased and the price paid. When the manufacturer adopts the BTS platform, consumers can access true quality information and choose to buy authentic products or counterfeits at a lower price (i.e., $p_M^B < p_M^{PE}, p_C^B < p_M^{PE}$). Therefore, adopting the BTS platform always increases the expected consumer surplus. Correspondingly, the expected social welfare will always be improved. Proposition 6 also shows that under a lower government combating level, adopting the BTS platform can improve the expected consumer surplus and the expected social welfare to a greater extent. The reason is explained as follows. Under weaker government enforcement, when the manufacturer adopts the BTS platform, the prices of two types of products will decrease to a greater extent (i.e., $\frac{\partial(p_M^{PE}-p_M^B)}{\partial e} < 0$, $\frac{\partial(p_M^{PE}-p_C^B)}{\partial e} < 0$). Therefore, the expected consumer surplus is improved to a greater extent. Furthermore, since the quality disclosure effect will strengthen as government enforcement weakens, the manufacturer will benefit more from adopting the BTS platform. Therefore, the expected social welfare is also improved to a greater extent.

Summarizing Proposition 4 and Proposition 6, we can know that adopting the BTS platform may increase the expected demand for counterfeits but always increases the expected consumer surplus. Therefore, our paper provides an important managerial insight. Specifically, the role of the BTS platform in improving consumer rights is stronger than that in combating counterfeits. In addition, the improving effect (i.e., the expected demand for authentic products, the expected consumer surplus, and the expected social welfare) of the BTS platform will also strengthen as government enforcement weakens.

Fig. 3 illustrates the impact of adopting the BTS platform on the expected demands for authentic products and counterfeits in cases where $q = 0.5$or 0.9. On the one hand, Fig. 3 shows that regardless of the quality of the counterfeit, adopting the BTS platform always increases the expected demand for authentic products, and the improvement level in demand for authentic products when $e = 0$ is greater than when $e = 0.05$. On the other hand, Fig. 3 shows that adopting the BTS platform decreases the expected demand for counterfeits only when $e = 0$ and $\lambda > \frac{45}{46}$. Fig. 4 illustrates the impact of adopting the BTS platform on the counterfeiter's expected profit, the expected consumer surplus, and the expected social welfare under the same parameters. Fig. 4 shows that adopting the BTS platform always decreases the counterfeiter's expected profit, and increases both the expected consumer surplus, and the expected social welfare, and the two indicators above change more when $e = 0$. The numerical results above show that adopting the BTS platform can effectively decrease the counterfeiter's expected profit while increasing the expected demand for authentic products, the expected consumer surplus, and the expected social welfare, but it may stimulate the expected demand for counterfeits. Moreover, the effect of adopting the BTS platform will improve when government enforcement weakens.

6. Robustness check

6.1. Asymmetric production cost

In this subsection, we examine the robustness of our key conclusions by relaxing the assumption about production cost. In the basic model, we assume that the manufacturer and the counterfeiter have the same production cost (i.e., the unit production costs of both the

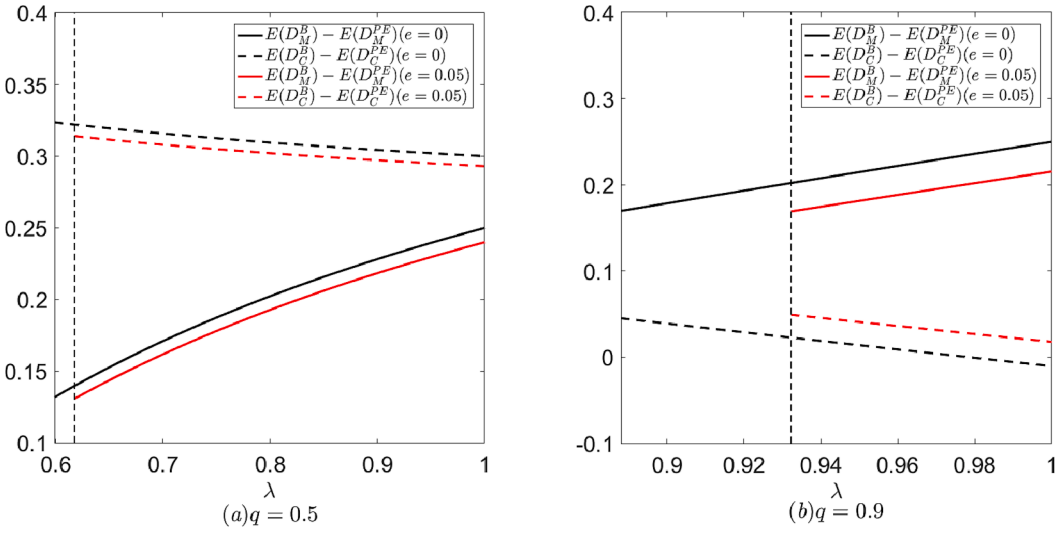


Fig. 3. The impact of adopting the BTS platform on the expected demands for authentic products and counterfeits.

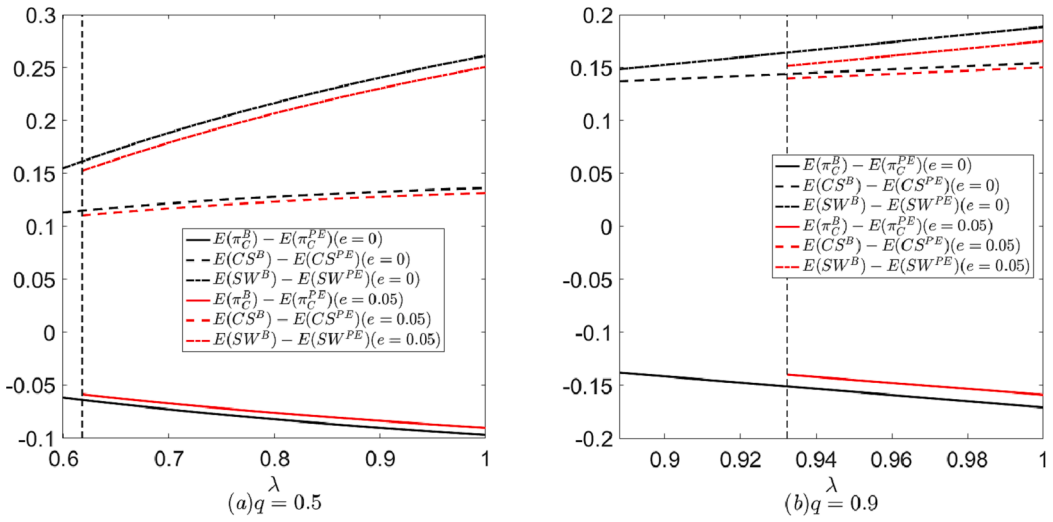


Fig. 4. The impact of adopting the BTS platform on the counterfeiter's expected profit, the expected consumer surplus and the expected social welfare.

authentic product and the counterfeit are normalized to zero). Below, we consider that the manufacturer and the counterfeiter have an asymmetric production cost. For simplicity, we assume the unit production cost of the counterfeit is 0, while the unit production cost of the authentic product is c. To ensure that both the manufacturer's expected profits with and without the BTS platform are non-negative, we assume $0 < c < \min[\frac{4(1-q)}{4-3q}, \frac{2(1-q)}{2-q-eq}]$.

Fig. 5 numerically examines the impact of deceptive sales under asymmetric production cost. We can find that regardless of cost differences (i.e., $c = 0.01$ or $c = 0.05$) and the quality of counterfeits (i.e., $q = 0.5$ or $c = 0.9$), deceptive sales may the expected demand for authentic products, the manufacturer's expected profit, the expected consumer surplus, and the expected social welfare simultaneously. Therefore, Proposition 1 still holds under asymmetric production cost. In addition, Fig. 5 also shows that an increase in the cost differences between counterfeits and authentic products will provide an incentive for the counterfeiter to implement non-deceptive sales in more cases. This is because cost differences result in a disadvantage for the manufacturer in price competition with the counterfeiter, the counterfeiter is more likely to implement the low-price strategy to maximize its expected profit. Fig. 6 and Fig. 7 numerically examine the effect of adopting the BTS platform under asymmetric production cost. We can find that the main results (i.e., Proposition 4-6) still hold under asymmetric production cost. That is, adopting the BTS platform always increases the expected demand for authentic products, the expected consumer surplus, and the expected social welfare while decreasing the counterfeiter's expected profit. However, the disadvantage is that it may increase the expected demand for counterfeits. In addition,

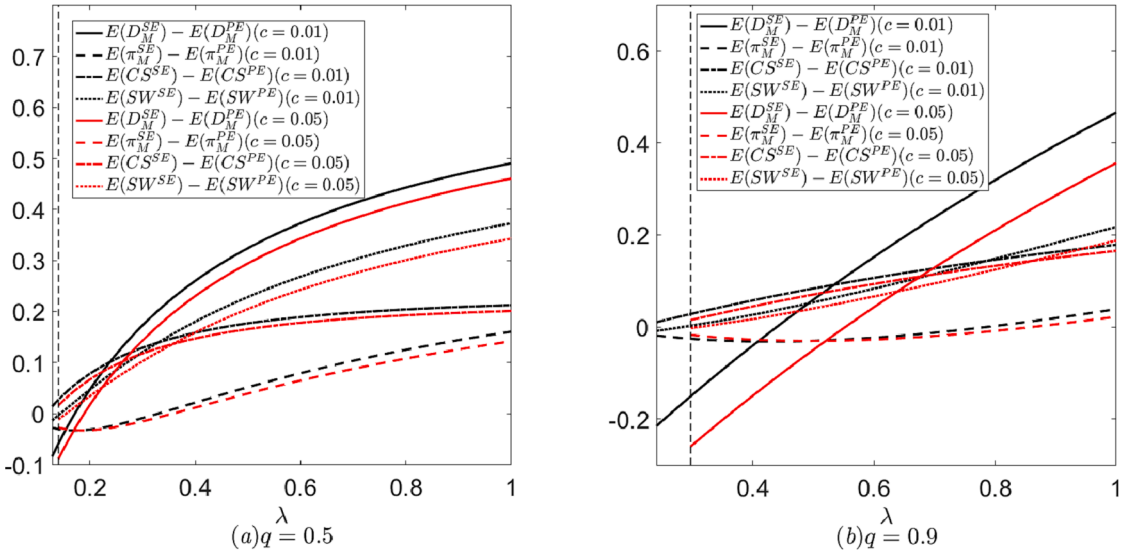


Fig. 5. The impact of deceptive sales under asymmetric production cost ($e = 0.01$).

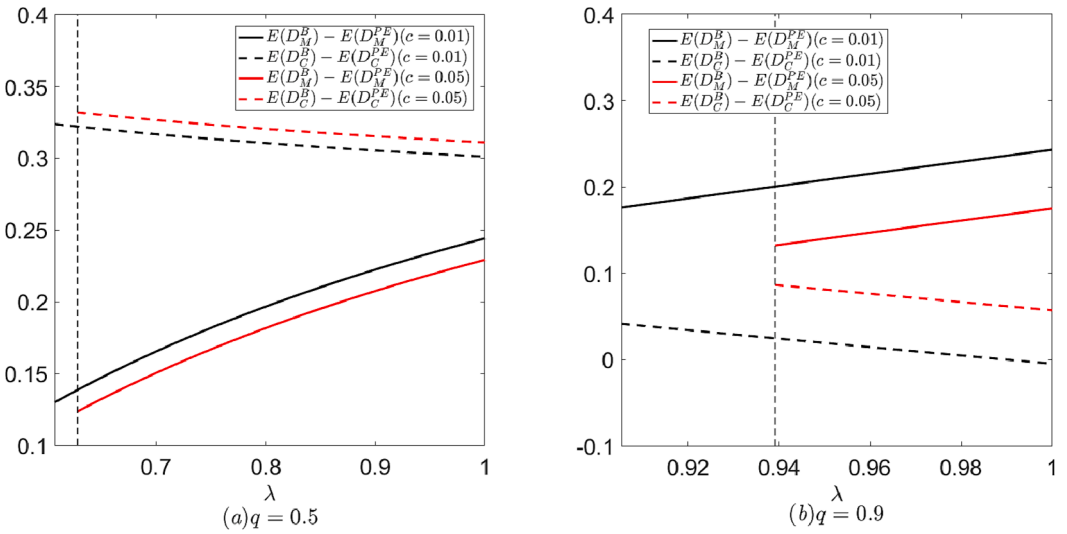


Fig. 6. The impact of adopting the BTS platform on the expected demands for authentic products and counterfeits under asymmetric production cost ($e = 0.01$).

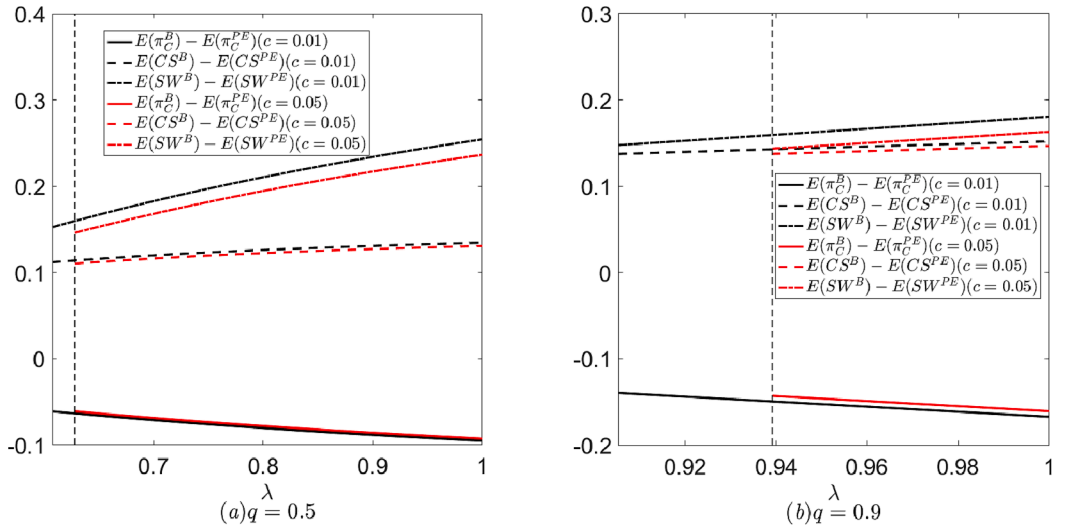


Fig. 7. The impact of adopting the BTS platform on the counterfeiter's expected profit, the expected consumer surplus and the expected social welfare under asymmetric production cost ($e = 0.01$).

the effect of adopting the BTS platform will improve under a lower production cost difference between counterfeits and authentic products.

6.2. *Asymmetric possibility of the counterfeiter being expelled*

In the basic model, we assume that the probability of the counterfeiter being expelled from the market by the government under both non-deceptive sales and deceptive sales is *e*. Below, we consider that under deceptive sales, the counterfeiter is less likely to be expelled by the government. Specifically, we assume that the possibility of the counterfeiter being expelled under deceptive sales is *ek*, where $k \in (0,1)$. The reduction of *k* means that the counterfeiter is more unlikely to be expelled by the government when implementing deceptive sales. Correspondingly, the expected profits of the manufacturer and the counterfeiter under pooling equilibrium are as follows:

$$E(\pi_M^{PE}) = p_M(1 + ek\lambda - \lambda)(1 - p_M)\# \tag{9}$$

$$E(\pi_C^{PE}) = \lambda(1 - ke)(1 - p_M)\# \tag{10}$$

Fig. 8 numerically examines the impact of deceptive sales under the asymmetric possibility of the counterfeiter being expelled. Fig. 9 and Fig. 10 numerically examine the effect of adopting the BTS platform under the asymmetric possibility of the counterfeiter being expelled. We can find that the numerical results under $k = 0.9$ are similar to that under $k = 0.1$, and the main results (i.e., Proposition 1 and Proposition 4-6) still hold under the asymmetric possibility of the counterfeiter being expelled.

6.3. Consumers concerns about product quality under deceptive sales

In the basic model, we assume that under deceptive sales, consumers do not aware of the existence of counterfeits, and their perceived quality for both products is equal to the true quality of the authentic product. In this subsection, we relax this assumption and consider that under deceptive sales, consumers have concerns about product quality. Let $\overline{q} \in [q, 1]$ denote consumers' perceived quality for both products under deceptive sales. In such a relaxed assumption, the expected profits of the manufacturer and the counterfeiter under pooling equilibrium are as follows:

$$E(\pi_M^{PE}) = p_M e(1 - p_M) + p_M(1 - e)(1 - \frac{p_M}{\overline{q}}) \tag{11}$$

$$E(\pi_C^{PE}) = \lambda(1 - ke)(1 - \frac{p_M}{\overline{q}})\# \tag{12}$$

Under the scenario without the BTS platform, the impact of consumer concerns on the counterfeiter's sales strategy is shown in Proposition 7.

Proposition 7. *If the manufacturer does not adopt the BTS platform, consumer concerns will incentivize the counterfeiter to implement non-deceptive sales. However, even if consumer concerns are high enough (i.e., $\overline{q} \to q$), the counterfeiter always implements deceptive sales if $q \le \frac{3-\sqrt{5-4e}}{2(1+e)}$.*

When the counterfeiter implements deceptive sales, consumer concerns will lead to lower prices they are willing to pay, which is unfavorable for the counterfeiter. Thus, consumer concerns will incentivize the counterfeiter to implement non-deceptive sales. However, Proposition 7 reveals that when the quality of the counterfeit is lower than a certain threshold, the counterfeiter always implements deceptive sales. This result implies that even if consumers are cautious in shopping, they may inadvertently purchase counterfeits.

Fig. 11 numerically examines the impact of deceptive sales when consumers have concerns about product quality under deceptive sales. We can find that regardless of the rampancy level of counterfeits, deceptive sales always decrease the expected demand for authentic products, the manufacturer's expected profit, the expected consumer surplus and the expected social welfare. These numerical results imply that when consumers have concerns about product quality, the negative impact of deceptive sales will expand. Fig. 12 and Fig. 13 numerically examine the effect of adopting the BTS platform when consumers have concerns about product quality under deceptive sales. We can find that when consumers' concerns are relatively high and the quality of the counterfeits is relatively low (i.e., $\overline{q} = \frac{1+q}{2}$ and $q = 0.5$), adopting the BTS platform may increase the counterfeiter's expected profit. This is because the reduction of consumers' perceived quality will weaken the negative impact of the quality disclosure effect on the counterfeiter's expected profit. For all that, under the relaxed assumption, adopting the BTS platform still always increase the expected demand for authentic products, the expected consumer surplus and the expected social welfare. Therefore, these numerical results show that the

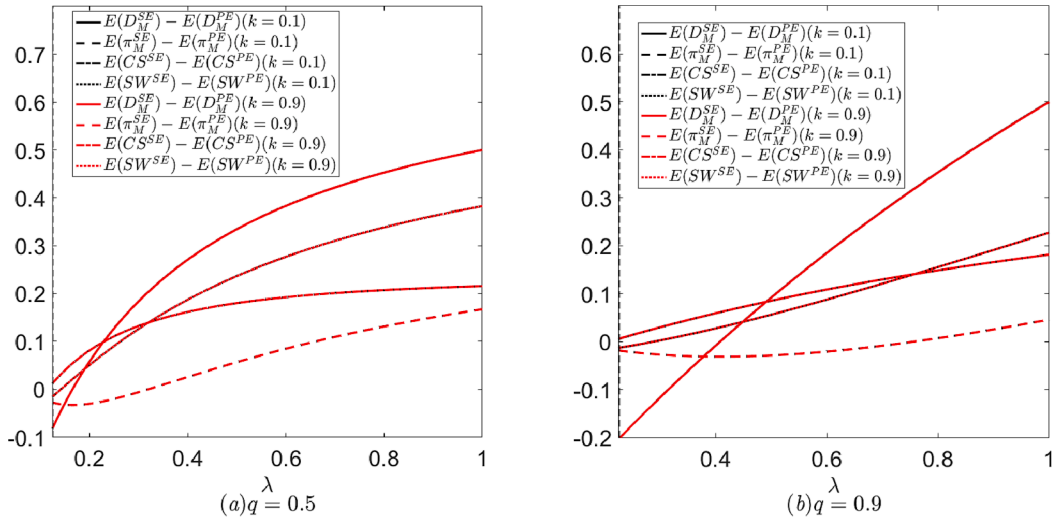


Fig. 8. The impact of deceptive sales under the asymmetric possibility of the counterfeiter being expelled ($e = 0.01$).

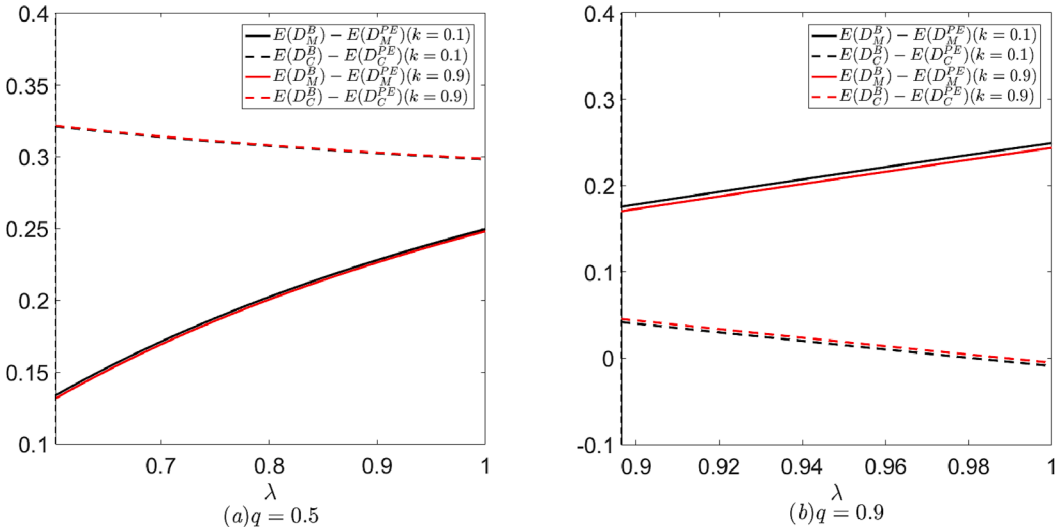


Fig. 9. The impact of adopting the BTS platform on the expected demands for authentic products and counterfeits under the asymmetric possibility of the counterfeiter being expelled ($e = 0.01$).

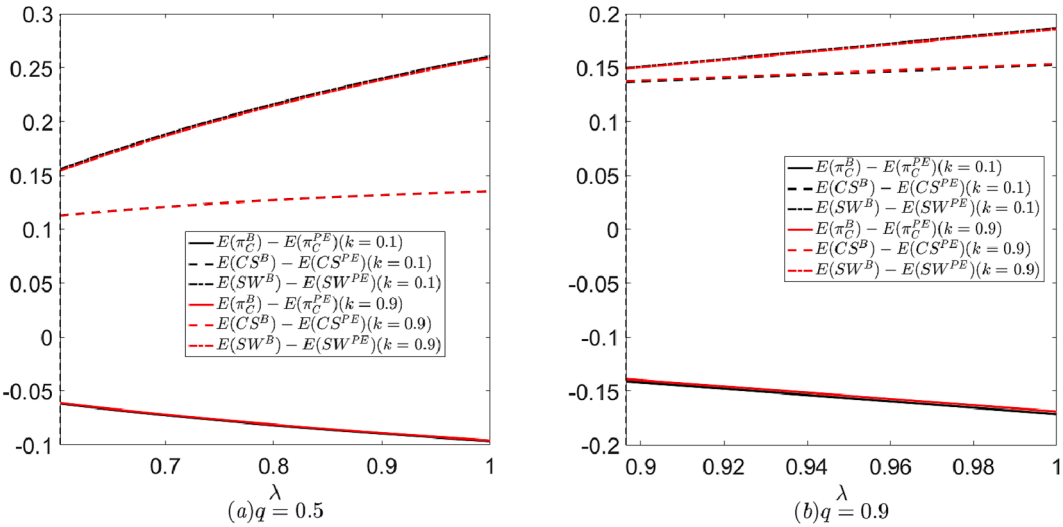


Fig. 10. The impact of adopting the BTS platform on the counterfeiter's expected profit, the expected consumer surplus and the expected social welfare under the asymmetric possibility of the counterfeiter being expelled ($e = 0.01$).

role of the BTS platform in improving consumer rights is stronger than that in combating counterfeits.

6.4. Self-implementing blockchain technology

In reality, there are also some brand-name firms dealing with counterfeits by self-implementing blockchain technology. In this subsection, we will examine the effect of self-implementing blockchain technology. Following Pun et al. (2021), we use c_B to denote the marginal cost of self-implementing blockchain technology. Correspondingly, when the manufacturer self-implements blockchain technology, his expected profit is as follows:

$$E(\pi_M^B) = (p_M - c_B)[(1-e)(1 - \frac{p_M - p_C}{1-q}) + e(1-p_M)] \tag{13}$$

Fig. 14 and Fig. 15 numerically examine the effect of self-implements blockchain technology. These numerical results show that our main results (i.e., Proposition 4-6) still hold.

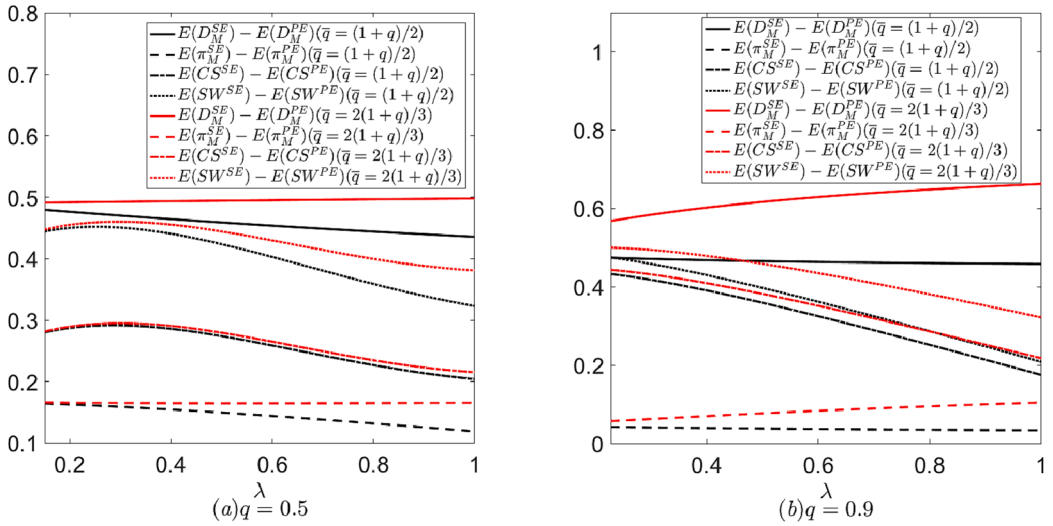


Fig. 11. The impact of deceptive sales when consumers have concerns about product quality under deceptive sales ($e = 0.01$).

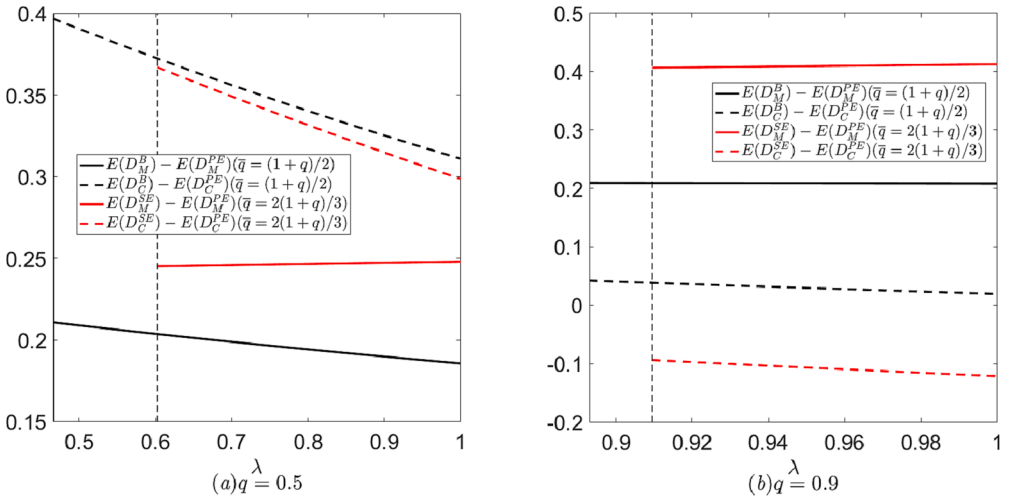


Fig. 12. The impact of adopting the BTS platform on the counterfeiter's expected profit, the expected consumer surplus and the expected social welfare when consumers have concerns about product quality under deceptive sales ($e = 0.01$).

7. Conclusions

This paper examines how the brand-name firm adopts the BTS platform for dealing with counterfeits under government enforcement. We consider a market consisting of a manufacturer (i.e., a brand-name firm) and a counterfeiter. The counterfeiter sells counterfeits with low quality but a high physical resemblance to authentic products. The manufacturer can choose whether to adopt the BTS platform. If the BTS platform is not adopted, the counterfeiter may implement deceptive sales. We develop a Stackelberg game model and a signaling game model under the scenarios with and without the BTS platform, respectively. We present several major findings and managerial insights as follows.

First, deceptive sales have a serious negative impact on the market. Under weak government enforcement and a high rampancy level of counterfeits, deceptive sales will decrease the expected demand for authentic products, the manufacturer's expected profit, the expected consumer surplus, and the expected social welfare. Although government enforcement can eliminate asymmetric quality information, it is not beneficial for consumers. Therefore, in order to improve the interests of itself and consumers, the manufacturer should consider adopting the BTS platform to solve the problem of asymmetric quality information.

Second, although adopting the BTS platform can eliminate asymmetric information, it is not always beneficial to the manufacturer. Whether the manufacturer should adopt the BTS platform depends on specific market conditions. However, weak government enforcement will incentivize the manufacturer to adopt the BTS platform.

Third, adopting the BTS platform can increase the expected demand for authentic products, the expected consumer surplus, and the

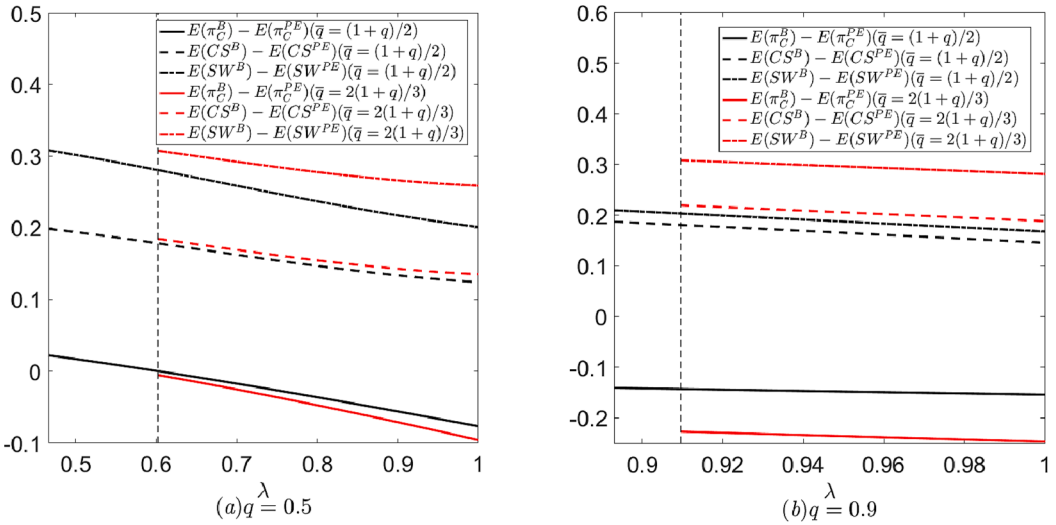


Fig. 13. The impact of adopting the BTS platform on the expected demands for authentic products and counterfeits when consumers have concerns about product quality under deceptive sales ($e = 0.01$).

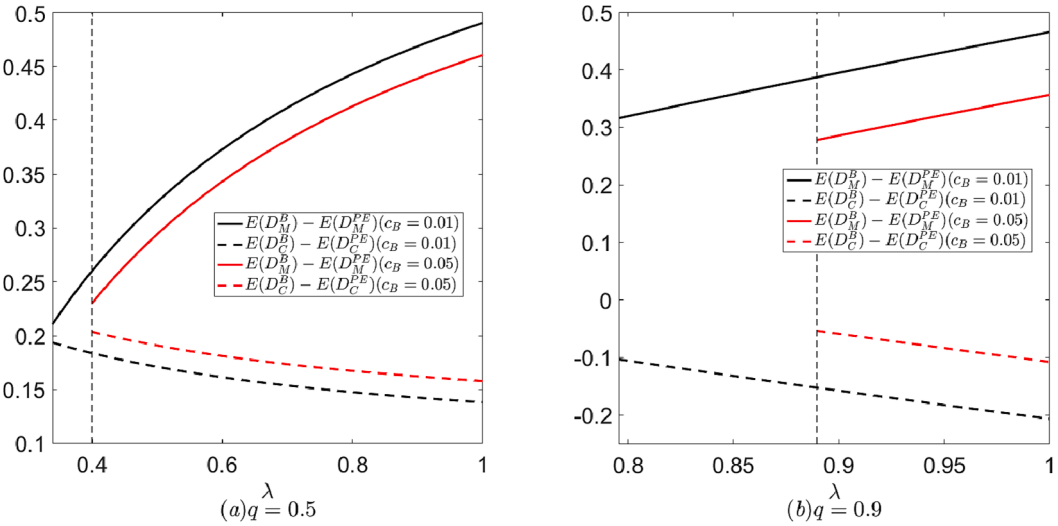


Fig. 14. The impact of adopting the BTS platform on the expected demands for authentic products and counterfeits when the manufacturer self-implements blockchain technology ($e = 0.01$).

expected social welfare, while decreasing the counterfeiter's expected profit. However, if the quality of the counterfeit is relatively low, adopting the BTS platform will also increase the expected demand for counterfeits. Therefore, the role of the BTS platform in improving consumer rights is stronger than that in combating counterfeits. In addition, both the improving effect and the combating effect of the BTS platform will strengthen as government enforcement weakens.

We point out some potential directions for future research. First, as observed in practice, there may be different types of counterfeits in the market for a certain brand name product (e.g., low-quality type and high-quality type). Therefore, it would be interesting to analyze how brand-name firms adopt the BTS platform in the market where there are different types of counterfeits. Second, our paper assumes that the government combating level is exogenously given. Another possible extension of our research would be to consider that the government combating level is endogenously determined based on welfare.

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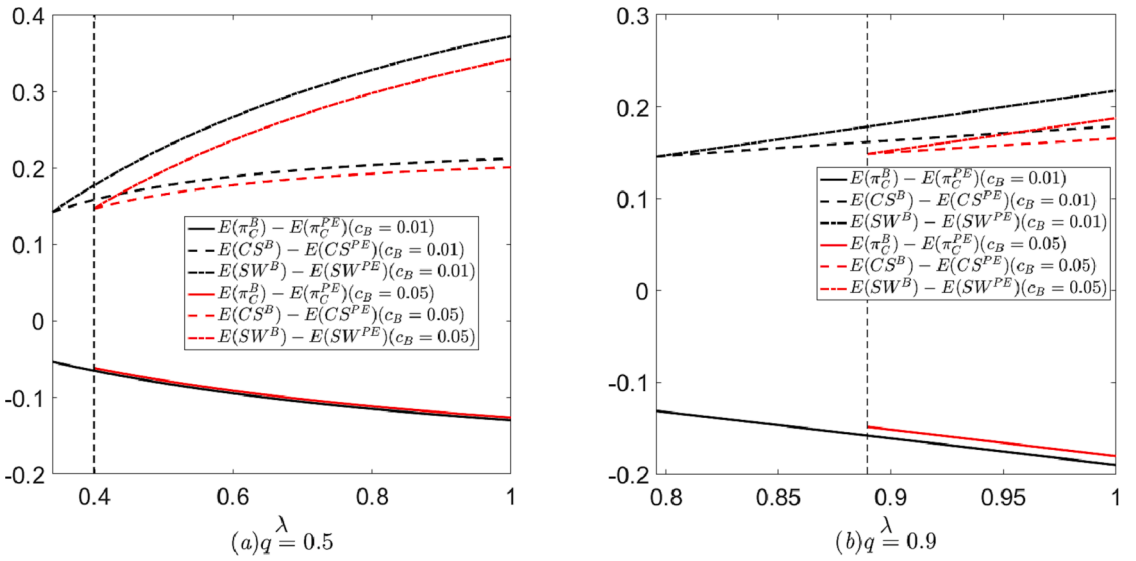


Fig. 15. The impact of adopting the BTS platform on the expected demands for authentic products and counterfeits when the manufacturer self-implements blockchain technology ($e = 0.01$).

CRediT authorship contribution statement

Chao He: Conceptualization, Methodology, Formal analysis, Writing – original draft. **Chunqiao Tan:** Conceptualization, Writing – original draft, Investigation, Funding acquisition. **W.H. Ip:** Visualization, Supervision. **C.H. Wu:** Software, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A

Proof of Lemma 1. By solving $\frac{\partial E(\pi_C^{SE})}{\partial p_C} = 0$, we can obtain $p_C^{SE}(p_M) = \frac{qp_M}{2}$. Substituting $p_C^{SE}(p_M)$ into Equation (1) and Equation (2), we can obtain $E(\pi_C^{SE}) = \frac{qp_M^2(1-e)}{4(1-q)}$ and $E(\pi_M^{SE}) = \frac{p_M(p_Mq-2q-2p_M+ep_Mq+2)}{2(1-q)}$. The constraint (5) can be expressed as $p_M > \frac{4\lambda(1-q)}{q+4\lambda(1-q)}$. By using KKT (Karush–Kuhn–Tucker) conditions, we can obtain three equilibrium outcomes as follows: (1) $p_M^{SE} = \frac{1-q}{2-q-eq}$ and $p_C^{SE} = \frac{q(1-q)}{2(2-q-eq)}$; (2) $p_M^{PE} = \frac{4\lambda(1-q)}{q+4\lambda-4\lambda q}$; (3) $p_M^{PE} = \frac{1}{2}$.

Below, we use intuitive criterion to refine three equilibrium outcomes. Before refinement, we define the out-of-equilibrium belief as follows: if the price of the counterfeit is the same as the authentic price, consumers cannot distinguish the counterfeit from the authentic product, otherwise, they judge that the higher-priced one is the authentic product and the lower-priced one is a counterfeit. Next, we begin to examine the stability of the three equilibrium outcomes under the above out-of-equilibrium belief.

We first examine the stability of the separating equilibrium outcome $p_M^{SE} = \frac{1-q}{2-q-eq}$ and $p_C^{SE} = \frac{q(1-q)}{2(2-q-eq)}$. If the counterfeiter sets a non-equilibrium price, the counterfeiter's expected profit must decrease due to a non-best response. Therefore, the separating equilibrium outcome $p_M^{SE} = \frac{1-q}{2-q-eq}$ and $p_C^{SE} = \frac{q(1-q)}{2(2-q-eq)}$ can survive under the intuitive criterion. Then, we examine the stability of the pooling equilibrium outcome $p_M^{PE} = \frac{4\lambda(1-q)}{q+4\lambda-4\lambda q}$. If the counterfeiter sets a non-equilibrium price, his expected profit under the best response is the same as under setting the equilibrium price, that is, $\max_{p_c} E[\pi_C(p_c \neq p_M^{PE})] = E[\pi_C(p_c = p_M^{PE})]$. Therefore, the counterfeiter has no incentive to set any non-equilibrium price. This result implies that the pooling equilibrium outcome $p_M^{PE} = \frac{4\lambda(1-q)}{q+4\lambda-4\lambda q}$ can survive under intuitive criterion. Finally, we examine the stability of the pooling equilibrium outcome $p_M^{PE} = \frac{1}{2}$. If the counterfeiter sets a non-equilibrium price, his expected profit may improve, that is, $\max_{p_c} E[\pi_C(p_c \neq p_M^{PE})] > E[\pi_C(p_c = p_M^{PE})]$ may hold true. Therefore, the pooling

equilibrium outcome $p_M^{PE}=\frac{1}{2}$ cannot survive under the intuitive criterion.

To sum up, the unique separating equilibrium outcome is $p_M^{SE}=\frac{1-q}{2-q-eq}$ and $p_C^{SE}=\frac{q(1-q)}{2(2-q-eq)}$; the unique pooling equilibrium outcome is $p_M^{PE}=\frac{4\lambda(1-q)}{q+4\lambda-4\lambda q}$. Substituting $p_M^{SE}=\frac{1-q}{2-q-eq}$ into constraint (5), we can obtain existing conditions for separating equilibrium: $\lambda<\frac{q}{4(1-eq)}=\bar{\lambda}(e,q)$.

Proof of Proposition 1. According to Lemma 1, we can obtain $E(D_M^{SE})=\frac{1}{2}$, $E(D_M^{PE})=\frac{q(1-\lambda+e\lambda)}{(q+4\lambda-4\lambda q)}$, $E(\pi_M^{SE})=\frac{1-q}{2(2-q-eq)}$, $E(\pi_M^{PE})=\frac{4q\lambda(1-q)(1-\lambda+e\lambda)}{(q+4\lambda-4\lambda q)^2}$, $E(CS^{SE})=\frac{4e^2+eq^2-9eq-q^2+q+4}{8(2-q-eq)^2}$, $E(CS^{PE})=\frac{q^2}{2(q+4\lambda-4\lambda q)^2}$, $E(SW^{SE})=\frac{4e^2q^2+5eq^2-13eq+3q^2-11q+12}{8(2-q-eq)^2}$ and $E(SW^{PE})=-\frac{q(8\lambda q-q-8\lambda-8e\lambda^2-8q\lambda^2+8\lambda^2+8eq\lambda^2)}{2(4\lambda+q-4\lambda q)^2}$.

First, we examine the impact of deceptive sales on the expected demand for authentic products. It is easy to derive that $\frac{\partial E(D_M^{PE})}{\partial\lambda}<0$. The minimum and maximum of $E(D_M^{PE})-E(D_M^{SE})$ are $\lim_{\lambda\to\bar{\lambda}}E(D_M^{PE})-E(D_M^{SE})=\frac{q(1-e)}{4(2-q-eq)}$ and $\lim_{\lambda\to1}E(D_M^{PE})-E(D_M^{SE})=\frac{3q+2eq-4}{2(4-3q)}$, respectively. Obviously, $\lim_{\lambda\to\bar{\lambda}}E(D_M^{PE})-E(D_M^{SE})\rangle0$. By solving $\lim_{\lambda\to1}E(D_M^{PE})-E(D_M^{SE})\le0$ and $E(D_M^{PE})-E(D_M^{SE})\langle0$, we can obtain $e\le\frac{4-3q}{2q}=e_1(q)$ and $\lambda>\frac{q}{2(2-q-eq)}=\lambda_1(e,q)$. Hence, we can conclude that deceptive sales will decrease the expected demand for authentic products if $e\le e_1(q)$ and $\lambda>\lambda_1(e,q)$. This ends the proof of Proposition 1(*i*).

Then, we examine the impact of deceptive sales on the manufacturer's expected profit. It is easy to derive that $E(\pi_M^{PE})$ first increases and then decreases in λ. The minimum and maximum of $E(\pi_M^{PE})-E(\pi_M^{SE})$ are $\lim_{\lambda\to\bar{\lambda}}E(\pi_M^{PE})-E(\pi_M^{SE})=\frac{q(1-q)(1-e)}{4(2-q-eq)^2}$ and $\lim_{\lambda\to1}E(\pi_M^{PE})-E(\pi_M^{SE})=-\frac{(1-q)(8e^2q^2+8eq^2-16eq+9q^2-24q+16)}{2(2-q-eq)(4-3q)^2}$, respectively. Obviously, $\lim_{\lambda\to\bar{\lambda}}E(\pi_M^{PE})-E(\pi_M^{SE})\rangle0$. By solving $\lim_{\lambda\to1}E(\pi_M^{PE})\le E(\pi_M^{SE})$ and $E(\pi_M^{PE})\langle E(\pi_M^{SE})$, we can obtain $e\le\frac{4-2q-\sqrt{32q-16-14q^2}}{4q}=e_2(q)$ and $\lambda>\frac{2q-2eq^2+q\sqrt{2q(1-e)(2-q-eq)}}{4(e^2q^2-2eq+q^2-2q+2)}=\lambda_2(e,q)$. Hence, we can conclude that deceptive sales will decrease the manufacturer's expected profit if $e\le e_2(q)$ and $\lambda>\lambda_2(e,q)$. This ends the proof of Proposition 1(*ii*).

Next, we examine the impact of deceptive sales on the expected consumer surplus. Obviously, $\frac{\partial E(CS^{PE})}{\partial\lambda}<0$. By simplifying, $\lim_{\lambda\to\bar{\lambda}}E(CS^{PE})\langle E(CS^{SE})$ can be converted into $(1-q)(1-e)\rangle0$, which always holds true. Hence, we can conclude that deceptive sales always decrease the expected consumer surplus. According to $E(CS^{SE})$ and $E(CS^{PE})$, we can obtain $\frac{\partial[E(CS^{SE})-E(CS^{PE})]}{\partial e}=\frac{q(1-q)(3q+7eq-10)}{8(2-q-eq)^3}$. Obviously, $\frac{\partial[E(CS^{SE})-E(CS^{PE})]}{\partial e}<0$. This ends the proof of Proposition 1(*iii*).

Finally, we examine the impact of deceptive sales on the expected social welfare. It is easy to derive that $\frac{\partial E(SW^{PE})}{\partial\lambda}<0$. The minimum and maximum of $E(SW^{PE})-E(SW^{SE})$ are $\lim_{\lambda\to\bar{\lambda}}E(SW^{PE})-E(SW^{SE})=\frac{q(1-e)(1-q)}{8(2-q-eq)^2}$ and $\lim_{\lambda\to1}E(SW^{PE})-E(SW^{SE})=\frac{(1-q)(32e^3q^3+96e^2q^3-192e^2q^2+69eq^3-312eq^2+336eq+23q^3-132q^2+272q-192)}{8(2-q-eq)^2(4-3q)^2}$, respectively. Obviously, $\lim_{\lambda\to\bar{\lambda}}E(SW^{PE})-E(SW^{SE})\rangle0$. Due to complexity, we analyze the value of $\lim_{\lambda\to1}E(SW^{PE})-E(SW^{SE})$ by using function images. From Fig. A1, we can deduce that under a certain condition, the

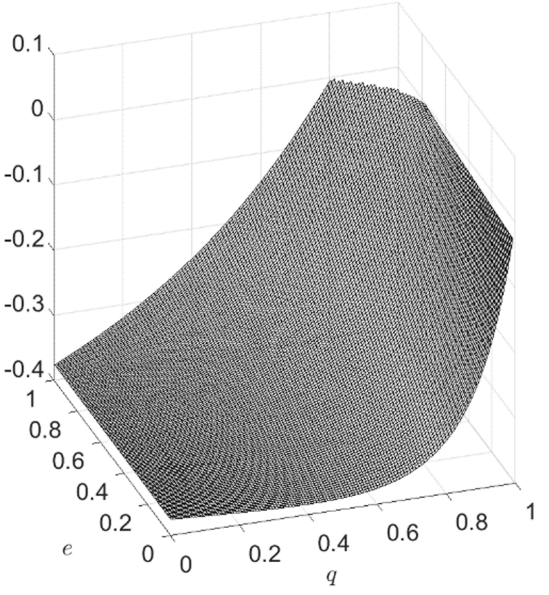


Fig. A1. The value of $\lim_{\lambda\to1}E(SW^{PE})-E(SW^{SE})$.

inequality $\lim_{\lambda\to 1} E(SW^{PE}) - E(SW^{SE})$ may hold. As shown in Fig. A2, when $e \le e_4(q)$, $\lim_{\lambda\to 1} E(SW^{PE}) - E(SW^{SE}) \le 0$. Hence, we can conclude that when $e \le e_4(q)$, there exists a threshold $\lambda > \lambda_3(e,q)$ such that $E(SW^{PE}) < E(SW^{SE})$ if $\lambda > \lambda_3(e,q)$, where $e_4(q)$ is the solution of $\lim_{\lambda\to 1} E(SW^{PE}) - E(SW^{SE}) = 0$with respect to e, and $\lambda_3(e,q)$ is the solution of $E(SW^{PE}) = E(SW^{SE})$with respect to λ. This ends the proof of Proposition 1(*iv*).

Proof of Proposition 2. According to Lemma 1, we can obtain $\frac{\partial \bar{\lambda}(e,q)}{\partial e} = \frac{q^2}{4(1-eq)^2}$ and $\frac{\partial E(CS^{SE})}{\partial e} = -\frac{q(1-q)(10-3q-7eq)}{8(2-q-eq)^3}$. Obviously, $\frac{\partial \bar{\lambda}(e,q)}{\partial e} > 0$ and $\frac{\partial E(CS^{SE})}{\partial e} < 0$. Hence, we can conclude that although a higher combating level forces the counterfeiter to implement non-deceptive sales, it will reduce the expected consumer surplus under symmetric quality information.

Proof of Lemma 2. By solving $\frac{\partial E(\pi_C^B)}{\partial p_C} = 0$, we can obtain $p_C(p_M) = \frac{qp_M}{2}$. Substituting $p_C(p_M)$ into Equation (7) and solving $\frac{\partial E(\pi_R^B)}{\partial p_M} = 0$, we can obtain $p_M(w) = \frac{2w-2q-qw-eqw+2}{2(2-q-eq)}$. Substituting $p_C(p_M)$ and $p_M(w)$ into Equation (6) and solving $\frac{\partial E(\pi_M^B)}{\partial w} = 0$, we can obtain $w = \frac{1-q}{(2-q-eq)}$. Correspondingly, we can obtain $p_M^B = \frac{3(1-q)}{2(2-q-eq)}$ and $p_C^B = \frac{3q(1-q)}{4(2-q-eq)}$.

Proof of Proposition 3. According to Lemma 2, we can obtain $E(\pi_M^B) = \frac{1-q}{4(2-q-eq)}$. By solving $E(\pi_M^B) \ge E(\pi_M^{PE})$ and $\lim_{\lambda\to 1} E(\pi_M^{PE}) \le E(\pi_M^B)$, we can obtain and $\lambda \ge \frac{q(3-q-2eq-\sqrt{(4-q-3eq)(2-q-eq)})}{4(1-eq)^2} = \lambda^*(e,q)$ and $e \le \frac{4-2q-\sqrt{q(8-5q)}}{4q} = e_4(q)$. Hence, we can conclude that the manufacturer can benefit from adopting the BTS platform if $\lambda \ge \lambda^*(e,q)$ and $e \le e_4(q)$. Due to $\lim_{\lambda\to\bar{\lambda}} E(\pi_M^{PE}) > E(\pi_M^{SE})$, we can deduce $\lambda^*(e,q) > \bar{\lambda}(e,q)$. Then, by determining first-order derivatives of $\lambda^*(e,q)$ with respect to e, we can obtain $\frac{\partial \lambda^*(e,q)}{\partial e} = \frac{q^2(11+6eq^2-12eq-10q+2q^2+3e^2q^2+2(2-q-eq)\sqrt{(4-q-3eq)(2-q-eq)})}{4(1-eq)^3\sqrt{(4-q-3eq)(2-q-eq)}}$. Due to complexity, we analyze the value of $\frac{\partial \lambda^*(e,q)}{\partial e}$ by using a function image. From Fig. A3, we can know $\frac{\partial \lambda^*(e,q)}{\partial e} > 0$.

Proof of Proposition 4. According to Lemma 1 and Lemma 2, we can obtain $E(D_M^B) = \frac{1}{4}$, $E(D_C^B) = \frac{3(1-e)}{4(2-q-eq)}$ and $E(D_C^{PE}) = \frac{q\lambda(1-e)}{(q+4\lambda-4\lambda q)}$. We first examine the impact of adopting the BTS platform on the expected demand for authentic products. Due to $\frac{\partial E(D_M^{PE})}{\partial \lambda} < 0$, we can know that the minimum of $E(D_M^B) - E(D_M^{PE})$ is $\lim_{\lambda\to\lambda^*} E(D_M^B) - E(D_M^{PE})$. Due to complexity, we analyze the value of $\lim_{\lambda\to\lambda^*(e,q)} E(D_M^B) - E(D_M^{PE})$ by using a function image. From Fig. A4, we can know $\lim_{\lambda\to\lambda^*(e,q)} E(D_M^B) - E(D_M^{PE}) > 0$, which implies $E(D_M^B) > E(D_M^{PE})$. According to $E(D_M^B)$ and $E(D_M^{PE})$, we can derive $\frac{\partial[E(D_M^B)-E(D_M^{PE})]}{\partial e} = -\frac{q\lambda}{4\lambda+q-4\lambda q}$. Obviously, $\frac{\partial[E(D_M^B)-E(D_M^{PE})]}{\partial e} < 0$. This ends the proof of Proposition 4(*i*).

Then, we examine the impact of adopting the BTS platform on the expected demand for counterfeits. It is easy to verify that $\frac{\partial E(D_C^{PE})}{\partial \lambda} < 0$. Therefore, we can know that the maximum of $E(D_C^{PE}) - E(D_C^B)$ is $\lim_{\lambda\to 1} E(D_C^{PE}) - E(D_C^B)$. Due to complexity, we analyze the value of $\lim_{\lambda\to 1} E(D_C^{PE}) - E(D_C^B)$ by using function images. From Fig. A5, we can deduce that under a certain condition, the inequality $\lim_{\lambda\to 1} E(D_C^{PE}) - E(D_C^B) < 0$ may hold. As shown in Fig. A6, when $e > e_4(q)$, $\lim_{\lambda\to 1} E(D_C^{PE}) - E(D_C^B) < 0$. By solving $\lim_{\lambda\to 1} E(D_C^{PE}) - E(D_C^B) = 0$, we can obtain $e_4(q) = \frac{17q-12-4q^2}{4q^2}$. Hence, we can conclude that adopting the BTS platform always increases the expected demand for

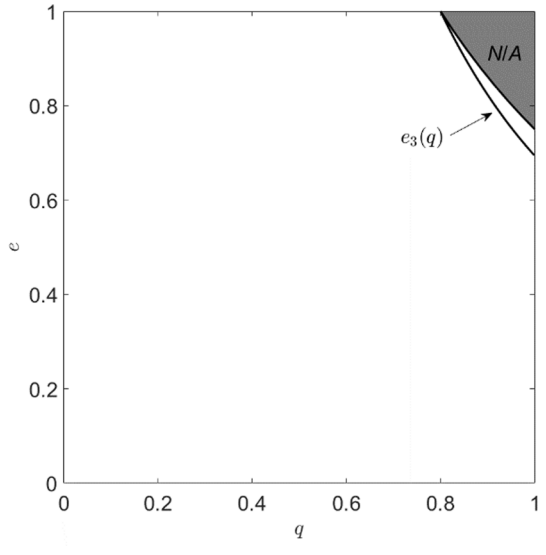


Fig. A2. The projection of $\lim_{\lambda\to 1} E(SW^{PE}) - E(SW^{SE}) > 0$ on the qe-plane ($e > e_3(q)$).

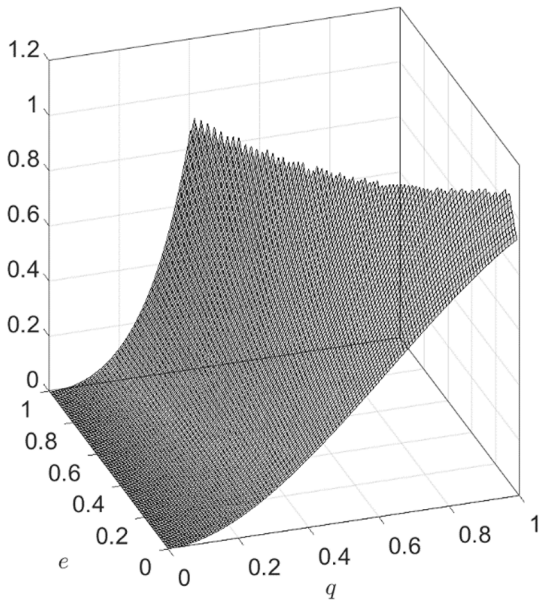


Fig. A3. The value of $\frac{\partial\lambda^*(e,q)}{\partial e}$.

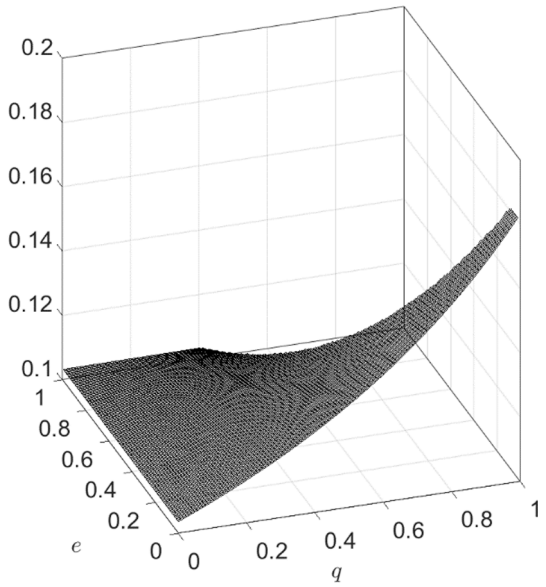


Fig. A4. The value of $\lim_{\lambda\to\lambda^*(e,q)} E(D_M^B) - E(D_M^{PE})$.

counterfeits if $q < \overline{q}(e) = \frac{17-\sqrt{97-192e}}{8(1+e)}$. From Fig. A4, it is easy to find that $\frac{\partial\overline{q}(e)}{\partial e} < 0$. This ends the proof of Proposition 4(*ii*).

Proof of Proposition 5. According to Lemma 1 and Lemma 2, we can obtain $E(\pi_C^{PE}) = \frac{4q\lambda^2(1-q)(1-e)}{(q+4\lambda-4\lambda q)^2}$ and $E(\pi_C^B) = \frac{9q(1-q)(1-e)}{16(2-q-eq)^2}$. Due to $\frac{\partial E(\pi_C^{PE})}{\partial\lambda} > 0$, we can know that the minimum of $E(\pi_C^{PE}) - E(\pi_C^B)$ is $\lim_{\lambda\to\lambda^*} E(\pi_C^{PE}) - E(\pi_C^B)$. Due to complexity, we analyze the value of $\lim_{\lambda\to\lambda^*(e,q)} E(\pi_C^{PE}) - E(\pi_C^B)$ by using a function image. From Fig. A7, we can know $\lim_{\lambda\to\lambda^*(e,q)} E(\pi_C^{PE}) - E(\pi_C^B) > 0$, which implies $E(\pi_C^B) \rangle E(\pi_C^{PE})$. Hence, we can conclude that adopting the BTS platform always decreases the counterfeiter's expected profit. According to $E(\pi_C^{PE})$ and $E(\pi_C^B)$, we can derive $\frac{\partial[E(\pi_C^{PE})-E(\pi_C^B)]}{\partial e} = q(1-q)(64e^3\lambda^2q^3 + 192e^2\lambda^2q^3 - 384e^2\lambda^2q^2 + 336e\lambda^2q^3 - 1056e\lambda^2q^2 + 912e\lambda^2q - 72e\lambda q^3 + 72e\lambda q^2 + 9eq^3 - 368\lambda^2q^3 + 768\lambda^2q^2 - 240\lambda^2q - 224\lambda^2 + 216\lambda q^3 - 360\lambda q^2 + 144\lambda q - 27q^3 + 18q^2 - 96q - 224)/16/(4\lambda + q - 4\lambda q)^2/(2 - q - eq)^3$. Let$\Delta_1 = 64e^3\lambda^2q^3 + 192e^2\lambda^2q^3 - 384e^2\lambda^2q^2 + 336e\lambda^2q^3 - 1056e\lambda^2q^2 + 912e\lambda^2q - 72e\lambda q^3 + 72e\lambda q^2 + 9eq^3 - 368\lambda^2q^3 +$

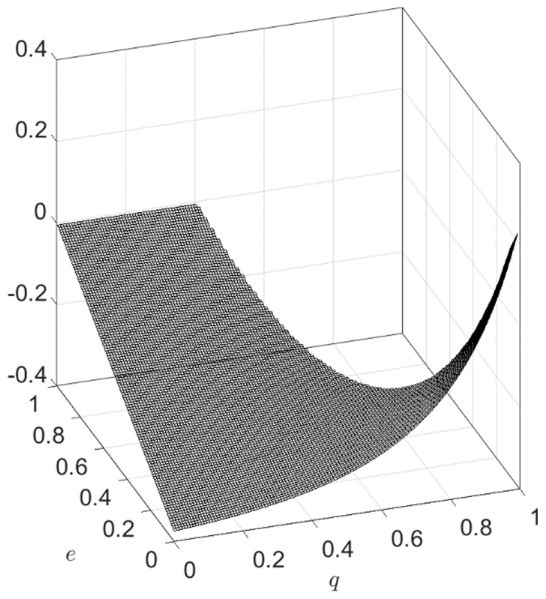


Fig. A5. The value of $\lim_{\lambda\to 1} E(D_C^{PE}) - E(D_C^B)$.

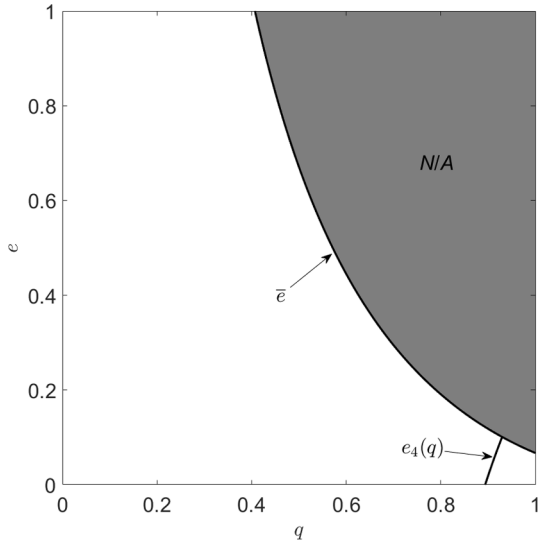


Fig. A6. The projection of $\lim_{\lambda\to 1} E(D_C^{PE}) - E(D_C^B) > 0$ on the qe-plane ($e > e_4(q)$).

$768\lambda^2q^2 - 240\lambda^2q - 224\lambda^2 + 216\lambda q^3 - 360\lambda q^2 + 144\lambda q - 27q^3 + 18q^2 - 96q - 224$. $\frac{\partial[E(\pi_C^{PE}) - E(\pi_C^B))]}{\partial e} < 0$ can be converted into $\Delta_1 < 0$. It is easy to deduce that $\frac{\partial \Delta_1}{\partial \lambda} < 0$. Hence, the maximum of Δ_1 is $\lim_{\lambda\to\lambda^*(e,q)} \Delta_1$. From Fig. A8, we can know that $\lim_{\lambda\to\lambda^*(e,q)} \Delta_1 < 0$, which implies that $\Delta_1 < 0$. Hence, we can conclude that $\frac{\partial[E(\pi_C^{PE}) - E(\pi_C^B))]}{\partial e} < 0$.

Proof of Proposition 6. According to Lemma 2, we can obtain $E(CS^B) = \frac{16e^2q^2 - 7eq^2 - 25eq - 5q^2 + 17q + 4}{32(2-q-eq)^2}$ and $E(SW^B) = \frac{16e^2q^2 + 5eq^2 - 37eq + 7q^2 - 19q + 28}{32(2-q-eq)^2}$. We first examine the impact of adopting the BTS platform on the expected consumer surplus. It is easy to deduce that the minimum of $E(CS^B) - E(CS^{PE})$ is $\lim_{\lambda\to\lambda^*(e,q)} E(CS^B) - E(CS^{PE})$. From Fig. A9, we can know $\lim_{\lambda\to\lambda^*(e,q)} E(CS^B) - E(CS^{PE}) \rangle 0$, which implies $E(CS^B) \rangle E(CS^{PE})$. Hence, we can conclude that adopting the BTS platform always increases the expected consumer surplus. According to $E(CS^B)$ and $E(CS^{PE})$, we can derive $\frac{\partial[E(CS^B) - E(CS^{PE})]}{\partial e} = \frac{3q(1-q)(q+13eq-14)}{32(2-q-eq)^3}$. Obviously, $\frac{\partial[E(CS^B) - E(CS^{PE})]}{\partial e} < 0$.

Then, we examine the impact of adopting the BTS platform on the expected social welfare. Since manufacturers will only adopt the

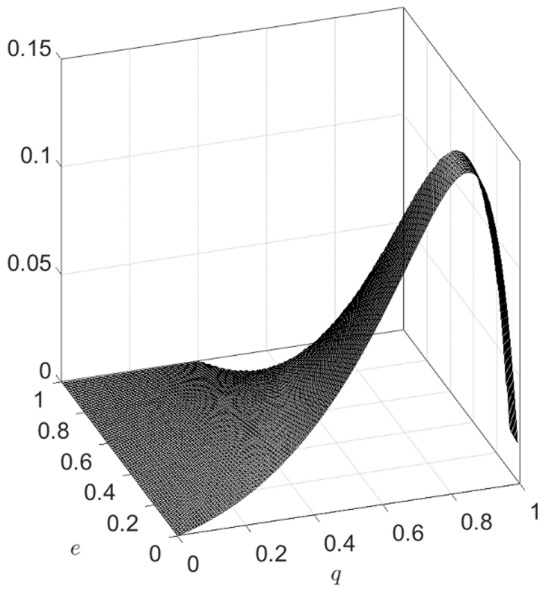


Fig. A7. The value of $\lim_{\lambda \to \lambda^*(e,q)} E(\pi_C^{PE}) - E(\pi_C^B)$.

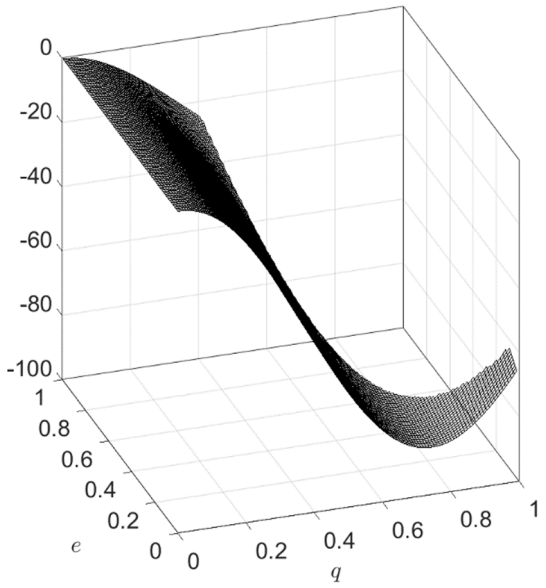


Fig. A8. The value of $\lim_{\lambda \to \lambda^*(e,q)} \Delta_1$.

BTS platform when they benefit from it and the expected consumer surplus always increases, the adoption of the BTS platform always increases the expected social welfare. According to $E(SW^B)$ and $E(SW^{PE})$, we can derive $\frac{\partial[E(SW^B)-E(SW^{PE})]}{\partial e} = \frac{q(1-q)\Delta_2}{(4\lambda+q-4\lambda q)^2(2-q-eq)^3}$, where$\Delta_2 = (128e^3\lambda^2q^3 + 384e^2\lambda^2q^3 - 768e^2\lambda^2q^2 + 816e\lambda^2q^3 - 2400e\lambda^2q^2 + 1968eq\lambda^2 - 216e\lambda q^3 + 216e\lambda q^2 + 27eq^3 - 16\lambda^2q^3 - 768\lambda^2q^2 + 1968q\lambda^2 - 1312\lambda^2 + 72\lambda q^3 + 72\lambda q^2 - 144q\lambda - 9q^3 - 18q^2)$. $\frac{\partial[E(SW^B)-E(SW^{PE})]}{\partial e} < 0$ can be converted into $\Delta_2 < 0$. It is easy to deduce that $\frac{\partial \Delta_2}{\partial \lambda} < 0$. Hence, the maximum of Δ_2 is $\lim_{\lambda \to \lambda^*(e,q)} \Delta_2$. From Fig. A10, we can know that $\lim_{\lambda \to \lambda^*(e,q)} \Delta_2 < 0$, which implies that $\Delta_2 < 0$. Hence, we can conclude that $\frac{\partial[E(SW^B)-E(SW^{PE})]}{\partial e} < 0$.

Proof of Proposition 6. Similar to the derivation process of Lemma 1, we can get the condition for the counterfeiter to implement deceptive sales (i.e., pooling equilibrium): $\lambda \geq \frac{q\overline{q}}{4(2\overline{q}+q-\overline{q}q-eq\overline{q}-1)} = \overline{\lambda}(e, q, \overline{q})$. By determining first-order derivatives of $\overline{\lambda}(e, q, \overline{q})$ with

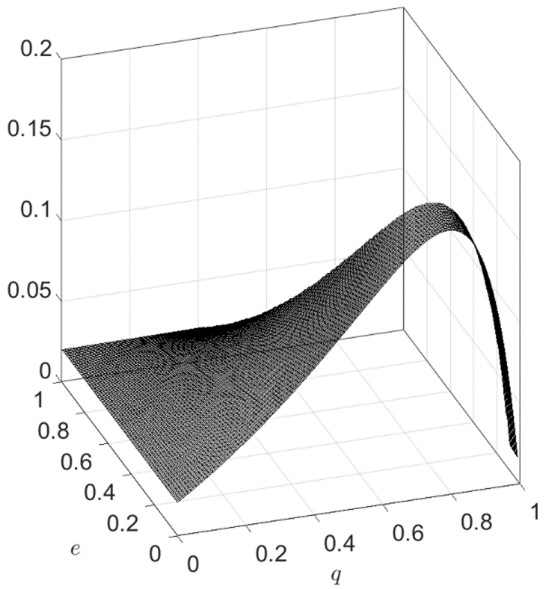


Fig. A9. The value of $\lim_{\lambda \to \lambda^{*}(e,q)} E\left(CS^{B}\right) - E(CS^{PE})$.

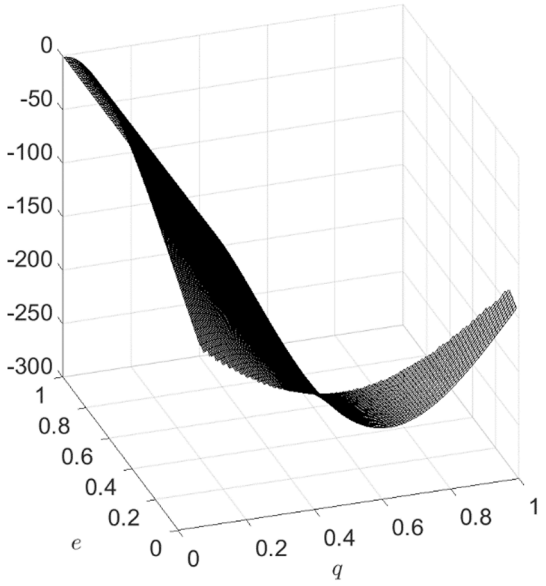


Fig. A10. The value of $\lim_{\lambda \to \lambda^{*}(e,q)} \Delta_2$.

respect to $\overline{q}$, we can obtain $\frac{\partial \overline{\lambda}(e,q,\overline{q})}{\partial \overline{q}} = \frac{q(q-1)}{4(q\overline{q}-2\overline{q}-q+eq\overline{q}+1)^2}$. Obviously, $\frac{\partial \overline{\lambda}(e,q,\overline{q})}{\partial \overline{q}} < 0$. Thus, if the manufacturer does not adopt the BTS platform, consumer concerns will incentivize the counterfeiter to implement non-deceptive sales. By solving $\lim_{\overline{q} \to q} \overline{\lambda}(e, q, \overline{q}) \leq 0$ with respect to q, we can obtain $q \leq \frac{3-\sqrt{5-4e}}{2(1+e)}$. Thus, even if consumer concerns are high enough, the counterfeiter always implements deceptive sales if $q \leq \frac{3-\sqrt{5-4e}}{2(1+e)}$.

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